MATH358 - Advanced Calculus

Differentiation, integration in $\mathbb{R}^n$

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Contents

1 Differentiation 2
1.1 Aside on Tangent Planes 7
1.2 Clairault's Theorem 7
1.3 Inverse Function Theorem 10
1.4 Taylor's Theorem in $\mathbb{R}^n$ and Lagrange Multipliers 14
1.5 Lagrange Multipliers 16
2 Riemann Integration 19
2.1 Integration over Box Domains in $\mathbb{R}^n$ 19
2.2 General Riemann Integration 21
2.3 Basic Properties of Integrals 23
2.4 Fubini's Theorem 23
2.4.1 Fubini for More General Domains 24
2.4.2 Fubini in $\mathbb{R}^n$ 25
2.4.3 Fubini's in $\mathbb{R}^3$ 26
2.5 Change of Variables Formula 27
3 Vector Fields, Curves and Surfaces 30
3.1 Integration along Curves 30
3.2 Integration of Vector Fields 30
3.2.1 Conservative Vector Fields 31
3.2.2 Poincaré Lemma 32
3.3 Poincaré Lemma 32
3.4 Integration on Surfaces 34
3.5 Green's Theorem and its Consequences 35
3.5.1 Harmonic Functions 39
3.6 Stoke's Theorem 39
3.7 Divergence Theorem 41
4 Practice Questions 43

§1 Differentiation

We say $\Omega \subset \mathbb{R}^n$ a *domain* if it is open and connected.

↪**Definition 1.1** (Differentiation): Let $f = (f_1, \ldots, f_m)^T : \Omega \to \mathbb{R}^m$, Ω a domain in $\mathbb{R}^n$ and $f_j : \Omega \to \mathbb{R}$. We say f *differentiable* at $x_0 \in \Omega$ if there exists a linear map $L : \mathbb{R}^n \to \mathbb{R}^m$ such that

$$\lim_{x \to x_0} \frac{\|f(x) - f(x_0) - L(x - x_0)\|}{\|x - x_0\|} = 0.$$

Remark 1.1: Note that the first norm on $\mathbb{R}^m$, the second on $\mathbb{R}^n$.

Remark 1.2: In terms of ε, δ, the definition says that $\forall\, \varepsilon > 0,\ \exists\, \delta > 0$ such that if $x \in \Omega \cap B(x_0, \delta)$, then $\|f(x) - f(x_0) - L(x - x_0)\| < \varepsilon \|x - x_0\|$.

↪**Theorem 1.1**: L as above is unique if it exists.

Proof. Suppose $L_1 \neq L_2$ both satisfy the definition. Then, for all $\varepsilon > 0$, there exists $\delta > 0$ such that if $0 < \|x - x_0\| < \delta$, then

$$\begin{aligned} \|(L_1 - L_2)(x - x_0)\| &\leq \|f(x) - f(x_0) - L_1(x - x_0)\| + \|f(x) - f(x_0) - L_2(x - x_0)\| \\ &\leq \varepsilon \|x - x_0\|, \end{aligned}$$

by differentiability (and the previous remark). In particular, $\|(L_1 - L_2)u\| < \varepsilon$ for all unit vectors u, which implies $\|(L_1 - L_2)u\| = 0$ and thus $L_1 = L_2$. ■

↪**Definition 1.2**: If f differentiable at x_0, we'll write $Df(x_0) = L$ for the *differential* of f at x_0.

↪**Proposition 1.1**: f differentiable at x_0 implies f continuous at x_0. In fact, f is Lipschitz at x_0.

Proof. Let $\delta > 0$ such that $\|x - x_0\| < \delta$ implies $\|f(x) - f(x_0) - Df(x_0)(x - x_0)\| < \|x - x_0\|$, which implies

$$\|f(x) - f(x_0)\| \leq \|Df(x_0)(x - x_0)\| + \|x - x_0\| \leq (\|L\| + 1)\|x - x_0\|,$$

which readily proves the statement. ■

↪**Proposition 1.2**: f differentiable at a point x_0 iff each of its component functions are differentiable at x_0.

↪**Definition 1.3**: For $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}^m$, define the *partial derivative*

$$\frac{\partial f_j}{\partial x_i}(x_1, \ldots, x_m) := \lim_{h \to 0} \frac{\left[f_j(x_1, \ldots, x_i + h, \ldots, x_m) - f_j(x_1, \ldots, x_i, \ldots, x_m)\right]}{h},$$

if the limit exists.

↪**Proposition 1.3**: Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}^m$ be differentiable at x_0. Then, $\frac{\partial f_j}{\partial x_i}(x_0)$ exists for each $i = 1, \ldots, n$ and $j = 1, \ldots, m$, and

$$L = Df(x_0) = \left(\frac{\partial f_j}{\partial x_i}(x_0) \right)_{\substack{1 \le i \le n \\ 1 \le j \le m}}.$$

We call this matrix the Jacobian or derivative of f at x_0.

Proof. Write $L = \left(a_{ji}\right)$ in the standard basis $e_1, \ldots, e_n$ for $\mathbb{R}^n$. Let $\varepsilon > 0$, fix some i with $1 \le i \le n$, and set $x := x_0 + he_i$, with $|h| < \delta$ sufficiently small. By differentiability,

$$\frac{\|f(x) - f(x_0) - L(x - x_0)\|}{\|x - x_0\|} = \left(\sum_{j=1}^{m} \left[\frac{f_j(x) - f_j(x_0)}{h} - a_{ji} \right]^2 \right)^{1/2}.$$

Since the limit as $h \to 0$ of the above ratio must be zero, the limit of each term in the summation as $h \to 0$ must be zero as well (being a sum of nonnegative terms), i.e.

$$\lim_{h \to 0} \frac{f_j(x) - f_j(x_0)}{h} = a_{ji} \quad \forall\, j = 1, \ldots, m.$$

But the limit on the left is just $\frac{\partial f_j}{\partial x_i}(x_0)$, which proves all of the claims in turn. ■

Remark 1.3: This proposition says that f differentiable at x_0 implies $\frac{\partial f_j}{\partial x_i}(x_0)$ exists for all i, j. The converse need not be true. Consider

$$f(x, y) := \begin{cases} \frac{xy}{\sqrt{x^2+y^2}} & (x, y) \neq 0 \\ 0 & \text{else} \end{cases}.$$

⊛ **Example 1.1**: Another counterexample as in the previous remark is the function

$$f(x,y) := \begin{cases} \frac{x^2y}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}.$$

Claim 1: f continuous at $(0,0)$. We have, for $(x,y) \neq (0,0)$,

$$\begin{aligned} |f(x,y) - f(0,0)| &= \left| \frac{x^2y}{x^2+y^2} \right| \\ &= \frac{x^2|y|}{x^2+y^2} \\ &\leq |y| \to 0 \text{ as } (x,y) \to (0,0), \end{aligned}$$

so we have continuity indeed.

Claim 2: $\partial_x f, \partial_y f$ exist at the origin, and are equal to zero. Note that $f(x,0) = 0$ for $x \neq 0$, and $f(0,0) = 0$, so it follows that $\partial_x f(0,0) = 0$. Similarly for $\partial_y f(0,0)$.

Claim 3: f is not differentiable at $(0,0)$. Suppose otherwise. Then, $L = Df(0,0) = (0,0)$, so

$$\begin{aligned} 0 &= \lim_{(x,y)\to(0,0)} \frac{|f(x,y) - f(0,0) - Df(0,0)(x,y)|}{\|(x,y)\|} \\ &= \lim_{(x,y)\to(0,0)} \frac{|f(x,y)|}{\|(x,y)\|} \\ &= \lim_{(x,y)\to(0,0)} \frac{x^2|y|}{(x^2+y^2)\cdot\sqrt{x^2+y^2}} \\ &= \lim_{(x,y)\to(0,0)} \frac{x^2|y|}{(x^2+y^2)^{3/2}}. \end{aligned}$$

Suppose $y = x$ in the final term (i.e., we approach the limit on a diagonal), and $x > 0$, then this ratio simplifies

$$\frac{x^3}{(2x^2)^{3/2}} = \frac{1}{2^{3/2}} \neq 0,$$

so we have a contradiction.

We can get a partial converse, however, if we assume continuity.

↪**Theorem 1.2**: Let $f = (f_1, \ldots, f_m) : \Omega \subset \mathbb{R}^n \to \mathbb{R}^m$. Suppose each $\frac{\partial f_j}{\partial x_i}$ is continuous at some $x^0 \in \Omega$. Then, f is differentiable at x^0.

Proof. We use MVT, and suppose $n = 2, m = 1$ for simplicity of notation, so that $f : \Omega \subset \mathbb{R}^2 \to \mathbb{R}$. We write $x = (x_1, x_2) \in \Omega, x^0 = \left(x_1^0, x_2^0\right)$. Let $\varepsilon > 0$. By assumption, there exists a $\delta > 0$ such that

$$\|y - x^0\| < \delta \Rightarrow \left|\frac{\partial f}{\partial x_i}(y) - \frac{\partial f}{\partial x_i}(x^0)\right| \leq \frac{\varepsilon}{2}, \quad i = 1, 2.$$

We write

$$\begin{aligned} f(x) - f(x^0) &= f(x_1, x_2) - f(x_1^0, x_2) + f(x_1^0, x_2) - f(x_1^0, x_2^0) \\ \text{(MVT, coordinate-wise)} \quad &= \frac{\partial f}{\partial x_1}(z_1, x_2)(x_1 - x_1^0) + \frac{\partial f}{\partial x_2}(x_1^0, z_2)(x_2 - x_2^0), \end{aligned}$$

for some z_1 between x_1 and x_1^0 and some z_2 between x_2 and x_2^0. Thus,

$$\begin{aligned} f(x) - f(x^0) - Df(x^0)(x - x^0) &= f(x) - f(x^0) - \left(\partial_{x_1} f(x^0), \partial_{x_2} f(x^0)\right) \cdot (x - x^0) \\ &= \frac{\partial f}{\partial x_1}(z_1, x_2)(x_1 - x_1^0) + \frac{\partial f}{\partial x_2}(x_1^0, z_2)(x_2 - x_2^0) \\ &\quad - \frac{\partial f}{\partial x_1}(x^0)(x_1 - x_1^0) - \frac{\partial f}{\partial x_2}(x^0)(x_2 - x_2^0) \\ &= \left[\partial_{x_1} f(z_1, x_2) - \partial_{x_1} f(x_1^0, x_2^0)\right](x_1 - x_1^0) \\ &\quad + \left[\partial_{x_2}(x_1^0, z_2) - \partial_{x_2} f(x_1^0, x_2^0)\right](x_2 - x_2^0). \end{aligned}$$

By choice of z_1, z_2 and for (x_1, x_2) in $B(x^0, \delta)$, we know $(z_1, x_2) \in B(x^0, \delta)$ and $(x_1^0, z_2) \in B(x^0, \delta)$ as well, so we can appeal to continuity. In addition, its clear that $|x_i - x_i^0| \leq \|x - x^0\|$. Thus, using continuity, we find

$$|f(x) - f(x^0) - Df(x^0)(x - x^0)| \leq \left(\frac{\varepsilon}{2} + \frac{\varepsilon}{2}\right)\|x - x^0\| = \varepsilon\|x - x^0\|,$$

so dividing both sides by $\|x - x^0\|$ immediately gives the result. ■

↪**Definition 1.4**: Suppose $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ has continuous $\frac{\partial f}{\partial x_i}$ at all points in Ω. Then, we say f is *continuously differentiable* (in Ω), and we write $f \in C^1(\Omega)$.

Remark 1.4: Continuity of partial derivatives is sufficient, but not necessary, for differentiability. For instance,

$$f(x, y) = \begin{cases} \frac{x^2 y^2}{x^2 + y^4} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}.$$

On readily computes $\partial_x f(0, 0) = \partial_y f(0, 0) = 0$, but along the parabola $x = t^2, y = t$ $(t \neq 0)$,

$$\partial_x f(t^2, t) = \frac{1}{2},$$

so $\partial_x f$ can't be continuous. However, f is still differentiable at $(0, 0)$: we claim $L = 0$, then

$$\frac{|f(x, y) - f(0, 0) - L(x, y)|}{\|(x, y)\|} = \frac{|f(x, y)|}{(x^2 + y^2)^{\frac{1}{2}}} = \frac{x^2 y^2}{(x^2 + y^4)(x^2 + y^2)^{\frac{1}{2}}} \leq \frac{y^2}{|x^2 + y^2|^{\frac{1}{2}}} \leq |y| \underset{(x,y) \to 0}{\to} 0.$$

↪**Proposition 1.4** (Basic Properties of Differentiation):

1. If $f, g : \Omega \to \mathbb{R}^m$ both differentiable at $x^0 \in \Omega$, then so is $F = f + g$, and
$$D(f+g)(x^0) = Df(x^0) + Dg(x^0).$$
2. If $f, g : \Omega \to \mathbb{R}^m$ both differentiable at $x^0 \in \Omega$, then so is $F = fg : \Omega \to \mathbb{R}$, and
$$DF(x^0) = f(x^0)Dg(x^0) + g(x^0)Df(x^0).$$
3. $f, g : \Omega \to \mathbb{R}$ both differentiable at x^0 with $g(x^0) \neq 0$, then so is $F = \frac{f}{g}$, and
$$DF(x^0) = \frac{DF(x^0)}{g(x^0)} - \frac{f(x^0)Dg(x^0)}{g^2(x^0)}.$$
4. (Chain Rule) Given $f : \Omega \subset \mathbb{R}^n \to \tilde{\Omega} \subset \mathbb{R}^m$ and $g : \tilde{\Omega} \to \mathbb{R}^k$, with f differentiable at x^0 and g differentiable at $y^0 = f(x^0)$, then $H = g \circ f : \Omega \subset \mathbb{R}^n \to \mathbb{R}^k$ is differentiable at x^0, and
$$DH(x^0) = Dg(y^0) \cdot Df(x^0),$$
in which one should read the "$\cdot$" as matrix multiplication.

Proof. 1., 2., 3. left as an exercise. We prove 4., the Chain Rule, for it is realistically the most interesting. Set $L := Dg(y_0) \cdot Df(x_0)$, and we'll write $y = f(x)$ (so in particular $y_0 = f(x_0)$, as in the statement). We need to show
$$\lim_{x \to x_0} \frac{\|H(x) - H(x_0) - L(x - x_0)\|}{\|x - x_0\|} = 0.$$
Let us work the numerator:
$$\begin{aligned} H(x) - H(x_0) - L(x - x_0) &= g(y) - g(y_0) - Dg(y_0)Df(x_0)(x - x_0) \\ &= g(y) - g(y_0) - Dg(y_0)(y - y_0) \\ &\quad + Dg(y_0)(y - y_0) - Dg(y_0)Df(x_0)(x - x_0) \\ &= g(y) - g(y_0) - Dg(y_0)(y - y_0) \\ &\quad + Dg(y_0)(f(x) - f(x_0) - Df(x_0)(x - x_0)). \end{aligned}$$
This means
$$\begin{aligned} \|H(x) - H(x_0) - L(x - x_0)\| \leq & \overbrace{\|g(y) - g(y_0) - Dg(y_0)(y - y_0)\|}^{=:(A)} \\ & + \|Dg(y_0)\| \overbrace{\|f(x) - f(x_0) - Df(x_0)(x - x_0)\|}^{=:(B)}. \end{aligned}$$
By differentiability of f at x_0, $(B) \to 0$ as $\|x - x_0\| \to 0$. We also have that, since f differentiable it is Lipschitz continuous, there is some $C > 0$ such that for $\|x - x_0\|$ sufficiently small,
$$(A) = \|y - y_0\| \cdot \frac{(A)}{\|y - y_0\|} \leq C\|x - x_0\| \frac{A}{\|y - y_0\|}.$$

By differentiability of g, the ratio $\frac{\|A\|}{\|y-y_0\|} \to 0$ as $\|y - y_0\| \to 0$. By continuity of f, $\|y - y_0\| = \|f(x) - f(x_0)\|$ will become small as $\|x - x_0\| \to 0$, so that we have in all $\frac{A}{\|x-x_0\|} \to 0$ as $\|x - x_0\| \to 0$. ■

Exercise 1.1: Let f differentiable in $\mathbb{R}^2$ and $g(r,\theta) := (r\cos\theta, r\sin\theta)$ with $(r,\theta) \in (0,\infty) \times [0, 2\pi)$. Let $F(r,\theta) = f(g(r,\theta))$. Compute $\frac{\partial F}{\partial \theta}$ and $\frac{\partial F}{\partial r}$.

§1.1 Aside on Tangent Planes

Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ differentiable on Ω. Then $Df(x) = \left(\frac{\partial f}{\partial x_1}(x), \ldots, \frac{\partial f}{\partial x_2}(x)\right) =: \nabla f(x)$, called the *gradient* of f. Let $S := \{(x,z) \in \Omega \times \mathbb{R} : z = f(x)\}$ be the *graph* of f. Then, for $x^0 \in \mathbb{R}$,

$$T_{x_0}S = \left\{(x,z) \in \mathbb{R}^n \times \mathbb{R} : z = f(x^0) + \sum_{j=1}^{n} \frac{\partial f}{\partial x_j}(x^0)\left(x_j - x_j^0\right)\right\}$$

is the *tangent plane* to S at x^0.

To see this, let $v \in \mathbb{R}^n$ be a unit vector and $x \in \Omega$. Define $g(t) := f(x + tv)$ for $f : \Omega \to \mathbb{R}$ differentiable (for t sufficiently small, $x + tv$ remains in Ω by openness). We find

$$g'(t) = \langle \nabla f(x + tv), v\rangle$$

for t sufficiently small.

↪**Proposition 1.5**: Suppose $\nabla f(x) \neq 0$. Then, $\nabla f(x)$ points in the direction of steepest increase of f.

Proof. For v a unit vector, the *directional derivative* in the direction of v is $D_v f(x) = \langle \nabla f(x), v\rangle = \|\nabla f(x)\|\cos(\theta)$ where θ the angle between $\nabla f(x)$ and v. This is maximized when $\theta = 0$, i.e. when $v = \frac{\nabla f(x_0)}{\|\nabla f(x_0)\|}$. ■

We can rewrite the graph S as the level set $\{(x,z) \in \Omega \times \mathbb{R} \mid g(x,z) = 0\}$ where $g(x,z) := z - f(x)$. Heuristically, $\nabla g(x_0, z_0)$ should be *normal* to the surface S at (x_0, z_0) (for steepest increase). As such, we define

$$T_{(x_0,z_0)}S := \{\nabla g(x_0,z_0) \cdot (x - x_0, z - z_0) = 0\}.$$

Note that

$$\nabla g(x_0,z_0) = \left(-\partial_{x_1} f(x_0), \ldots, -\partial_{x_n} f(x_0), 1\right),$$

so that

$$T_{(x_0,z_0)} = \{z - z_0 = \nabla f(x_0) \cdot (x - x_0)\},$$

which gives the definition from above.

§1.2 Clairault's Theorem

Here, the question is, given $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ twice differentiable, when can we exchange order of second-order partial derivatives, i.e. when is

$$\frac{\partial^2 f}{\partial x_j \partial x_i} = \frac{\partial^2 f}{\partial x_i \partial x_j}, \quad \forall\, i,j = 1,\dots,n?$$

We need to establish first a generalization of the mean-value theorem. First, note that if

$$\gamma : (a,b) \to \mathbb{R}^n, \quad g : \Omega \subset \mathbb{R}^n \to \mathbb{R}$$

are two differentiable functions with $\gamma((a,b)) \subset \Omega$, then by the chain rule, if we put $H(t) := g(\gamma(t))$,

$$\frac{\partial H}{\partial t} = Dg(\gamma(t)) \cdot D\gamma(t), \quad D\gamma(t) = (\gamma_1'(t),\dots,\gamma_n'(t)).$$

↪**Theorem 1.3** (Mean-Value Theorem): Let $B \subset \mathbb{R}^n$ be a ball and $f : B \to \mathbb{R}$ be differentiable for all $x \in B$. Then, for any $x, y \in B$, there exists $z \in B$ (in particular, we can find z which lies along the line segment between x and y) such that

$$f(x) - f(y) = Df(z) \cdot (x - y).$$

In particular, $|f(x) - f(y)| \leq \|Df(z)\|\|x - y\|$.

Proof. Let $x, y \in B$ fixed and let $\gamma(t) := tx + (1-t)y$ for $t \in [0,1]$. We see that $\gamma(t) \in B$ for all $t \in [0,1]$, and that $D\gamma(t) = x - y$. Set $F(t) := f(\gamma(t))$ (i.e., we restrict f to its values along the straight line along x and y), noting $F : \mathbb{R} \to \mathbb{R}$. So, by 1-dimensional mean-value theorem, there is some $t^* \in [0,1]$ such that

$$\begin{aligned} f(x) - f(y) = F(1) - F(0) = F'(t^*) &= Df\Big(\underbrace{t^*x + (1-t^*)y}_{=:z \in B}\Big) \cdot D\gamma(t) \\ &= Df(z) \cdot (x - y). \end{aligned}$$

■

Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}^m$ differentiable. Remember that $Df : \Omega \to \mathbb{R}^{m \times n}$.

↪**Definition 1.5**: We say f twice differentiable at x if Df exists locally to x and Df is differentiable at x. We write

$$D^2 f = D(Df),$$

and similarly

$$D^k f := D\big(D^{k-1} f\big)$$

with an analogous definition.

↪**Definition 1.6**: Given $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$, we see that $f \in C^k(\Omega)$ for $k \in \mathbb{Z}_+$ if all the partial derivatives to order k exist and are continuous in Ω.

↪**Definition 1.7**: If $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ twice differentiable, the *Hessian matrix* is given by

$$H_f(x) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_n} \end{pmatrix}.$$

Exercise 1.2: Let $f(x,y) := \begin{cases} \frac{(xy)(x^2-y^2)}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$ and compute $H_f(x,y)$.

↪**Theorem 1.4** (Clairault): Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ be twice differentiable at $x \in \Omega$. Then,

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(x) = \frac{\partial^2 f}{\partial x_j \partial x_i}(x), \quad \forall\, i,j = 1,\dots,n.$$

↪**Corollary 1.1**: If $\frac{\partial^2 f}{\partial x_i \partial x_j}$ are all continuous at $x \in \Omega$ for $i,j = 1,\dots,n$, then $\frac{\partial^2 f}{\partial x_i \partial x_j}(x) = \frac{\partial^2 f}{\partial x_j \partial x_i}(x)$.

Proof. (Of Clairault's) It's enough to consider $n = 2$. Fix $(x,y) \in \Omega$, and note that for $s,t \in \mathbb{R}$ sufficiently small, $(x+s, y+t) \in \Omega$ by openness. Set

$$\begin{aligned} \Delta(s,t) &:= f(x+s,y+t) - f(x,y+t) - f(x+s,y) + f(x,y) \\ &= g_t(x+s) - g_t(x), \quad g_t(u) := f(u,y+t) - f(u,y). \end{aligned}$$

By the mean-value theorem, there is some $\xi_{s,t}$ between x and $x+s$ suh that

$$\Delta(s,t) = \frac{\partial g_t}{\partial x}(\xi_{s,t}) \cdot s = \left[\frac{\partial f}{\partial x}(\xi_{s,t}, y+t) - \frac{\partial f}{\partial x}(\xi_{s,t}, y) \right] s. \quad (\ddagger)$$

By assumption, $\frac{\partial f}{\partial x}$ is differentiable at (x,y), so

$$\frac{\partial f}{\partial x}(z_1,z_2) = \frac{\partial f}{\partial x}(x,y)(z_1 - x) + \frac{\partial^2}{\partial x^2}(x,y)(z_2 - y) + E_1(z_1,z_2), \quad (\dagger)$$

where

$$\frac{|E_1(z_1,z_2)|}{\sqrt{(z_1-x)^2 + (z_2-y)^2}} \to 0, \quad \text{as } (z_1,z_2) \to (x,y).$$

Evaluating $(\dagger)$ at $(z_1,z_2) = (\xi_{s,t}, y+t)$ and $(\xi_{s,t}, y)$, and plugging into $(\ddagger)$ yields

$$\Delta(s,t) = \left(\frac{\partial^2 f}{\partial y \partial x}(x,y)t + E_1(\xi_{s,t}, y+t) - E_1(\xi_{s,t}, y) \right) s.$$

Let $s = t$ and let $t \to 0$. We claim that

$$\frac{\partial^2 f}{\partial y \partial x}(x,y) = \lim_{s=t\to 0} \frac{\Delta(s,t)}{st} = \frac{\partial^2 f}{\partial x \partial y}(x,y).$$

The first equality is obvious from the assumptions on the error terms. On the other hand, we can switch the order of the middle terms in $\Delta(s,t)$ and write

$$\begin{aligned}\Delta(s,t) &= f(x+s,y+t) - f(x,y+t) - f(x+s,y) + f(x,y)\\ &= h_s(y+t) - h_s(y), \quad h_s(u) := f(x+s,u) - f(x,u).\end{aligned}$$

Repeating the same argument as above with g_t, we get that

$$\Delta(s,t) = \left(\frac{\partial^2}{\partial x \partial y}(x,y)s + E_2(x+s,\eta_{s,t}) - E_2(x,\eta_{s,t})\right)t,$$

where $\eta_{s,t}$ lies between y and $y+t$, and

$$|E_2(x+s,\eta_{s,t})| \leq |s^2+t^2|, \quad |E_2(x,\eta_{s,t})| \leq \sqrt{s^2+t^2}.$$

Setting $s=t$ here, we get

$$\lim_{\substack{s,t\to 0\\ s=t}} \frac{\Delta(s,t)}{st} = \frac{\partial^2}{\partial x \partial y}(x,y).$$

This proves the claim. ■

§1.3 Inverse Function Theorem

↪**Theorem 1.5** (In 1D): If $f:(a,b)\to(c,d)$ is differentiable with $f'(x)>0$, then there exists $g:(c,d)\to(a,b)$ differentiable such that $y=f(x) \Leftrightarrow x=g(y)$ (i.e. $x=g(y)$).

In higher dimensions, we recall some preliminaries before proving.

↪**Theorem 1.6**: Let (X,d) a complete metric space and $f:X\to X$ a contraction mapping, with $d(f(x_2),f(x_1)) \leq \alpha d(x,y)$ for all $x,y\in X$ for some $0<\alpha<1$. Then, there exists a unique $x_0\in X$ such that $f(x_0)=x_0$.

We will write $M_n := \{n\times n \text{ matrices}\} \cong \mathbb{R}^{n^2}$, and $\|A\| := \sqrt{\sum_{i,j=1}^n a_{ij}^2}$ where $A := \left(a_{ij}\right) \in M_n$. We use

$$\mathrm{GL}(n) := \{A\in M_n : \det(A)\neq 0\} = \det^{-1}(\mathbb{R}\setminus\{0\}), \quad \det : M_n \to \mathbb{R}.$$

Remark that since $\mathbb{R}\setminus\{0\}$ is open, and the map det is continuous (it can be written as a polynomial in the entries a_{ij}'s of the matrix A), we know that $\mathrm{GL}(n)$ an open subset of M_n.

Consider the map

$$f:\mathrm{GL}(n)\to\mathrm{GL}(n), \quad f(A) := A^{-1}.$$

↪**Lemma 1.1**: $\mathrm{GL}(n)\subset M_n$ open and $f\in C^k$ for all $k=1,2,\ldots$.

Proof. We already proved the first statement in our remarks above.

Let $A(j|i)$ be $(n-1)\times(n-1)$ matrix with its jth row and ith columns deleted, then recall

$$\operatorname{adj}(A) = \big((-1)^{i+j} \det A(j|i)\big).$$

By Cramer's formula from linear algebra,

$$f(A) = A^{-1} \frac{1}{\det(A)} \operatorname{adj}(A),$$

which is in C^k since $\det(A)$ C^k (being a polynomial in the coefficients of A) and $\det(A) \neq 0$. ■

↪**Theorem 1.7** (Inverse Function Theorem): Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}^n$ be C^1. Let $x_0 \in \Omega$ and assume $Df(x_0) \in \mathrm{GL}(n)$. Then, there exist domains U and V of x_0 and $f(x_0)$ resp. such that $f(U) = V$ and $f|_U$ has a C^1 inverse map $f^{-1} : V \to U$. Moreover, for any $y \in V$ and $x = f^{-1}(y)$, $Df^{-1}(y) = [Df(x)]^{-1}$.

Remark 1.5: By the first lemma above, if $f \in C^k$, $k \geq 1$, we get the same regularity for f^{-1}.

Proof. By translation, its enough to assume $x_0 = f(x_0) = y_0 = 0$ and $Df(x_0) = \mathrm{Id}$ by replacing f with $[Df(0)]^{-1}f$, so we have a mapping

$$f : \Omega \to \mathbb{R}^n, \quad f(0) = 0, Df(0) = \mathrm{Id}.$$

Fix $y \in V$ and set

$$g_y(x) := y + x - f(x),$$

remark that

$$g_y(x) = x \Leftrightarrow y = f(x),$$

so it suffices to show g_y as a mapping $\mathbb{R}^n \to \mathbb{R}^n$ is a contraction mapping, and

$$Dg_y(0) = \mathrm{Id} - \mathrm{Id} = 0.$$

If $f \in C^1(U)$, then $g_y \in C^1(U)$ so that $Dg_y \in C^0(U)$ (similar if $f \in C^k \Rightarrow g_y \in C^k$). Since $Dg_0 \in C^0(U)$, there exists some $\delta > 0$ sufficiently small such that $\|Dg_0(x)\| \leq \frac{1}{2}$, for all $x \in B_\delta(0)$. By mean-value theorem, there exists some $z \in B_\delta(0)$ such that

$$\begin{aligned}
\|g_0(x)\| &= \left\| g_0(x) - \underbrace{g_0(0)}_{=0} \right\| \\
&\leq \|Dg_0(z)\| \|x\| \\
&\leq \frac{\|x\|}{2} < \frac{\delta}{2},
\end{aligned}$$

which implies we can view

$$g_0 : B_\delta(0) \to B_{\delta/2}(0).$$

It follows that

$$g_y : B_\delta(0) \to B_\delta(0), \quad \forall\, y \in B_{\delta/2}(0),$$

using the fact $g_y = y + g_0$ and the triangle inequality. By MVT once again for any $x, x' \in B_\delta(0)$, there exists $y \in B_{\delta/2}(0)$ such that

$$\begin{aligned}\left\|g_y(x) - g_y(x')\right\| &= \|g_0(x) - g_0(x')\| \\ &\leq \|Dg_0(y)\|\|x - x'\| \\ &\leq \frac{\|x - x'\|}{2}\end{aligned}$$

hence $g_y : B_\delta \to B_\delta$ is a contraction mapping. By the fixed-point theorem, there exists a unique point $x \in B_\delta(0)$ such that $g_y(x) = x \Leftrightarrow y = f(x)$. That is, there exists an inverse map $f^{-1} : B_{\delta/2}(0) \to B_\delta(0)$. Moreover, for any $x, x' \in B_\delta(0)$,

$$\begin{aligned}\|x - x'\| &\leq \|f(x) - f(x')\| + \|g_0(x) - g_0(x)\| \\ &\leq \|f(x) - f(x')\| + \frac{1}{2}\|x - x'\|,\end{aligned}$$

i.e.

$$\|x - x'\| \leq 2\|f(x) - f(x')\|.$$

From here, we know that for $y, y' \in B_{\delta/2}(0)$,

$$\left\|f^{-1}(y) - f^{-1}(y')\right\| \leq 2\|y - y'\| \Rightarrow f^{-1} \in C^0\big(B_{\delta/2}(0)\big).$$

Next, we need to show that $Df^{-1}(y)$ exists for $y \in B_{\delta/2}(0)$ for small $\delta > 0$. Since $Df(0) \in \mathrm{GL}(n)$, we know $Df(x) \in \mathrm{GL}(n)$ if $x \in B_\delta(0)$ (possible after shrinking $\delta > 0$). Set

$$W := f^{-1}\big(B_{\delta/2}(0)\big),$$

and choose $R > 0$ suff. small so that

$$\overline{B_R(0)} \subset W.$$

Since $[Df]^{-1} \in C^0\big(\overline{B_R}\big)$ and $\overline{B_R}(0)$ is compact,

$$\left\|[Df(x)]^{-1}\right\| \leq K, x \in \overline{B_r(0)}.$$

Then, given $y, y' \in B_{\delta/2}(0)$ and with $x = f^{-1}(y), x' = f^{-1}(y')$, we find

$$\begin{aligned}\frac{\left\|f^{-1}(y) - f^{-1}(y') - [Df(x')]^{-1}(y - y')\right\|}{\|y - y'\|} &= \frac{\left\|x - x' - [Df(x')]^{-1}(f(x) - f(x'))\right\|}{\|f(x) - f(x')\|} \\ &= \frac{\|x - x'\|}{\|f(x) - f(x')\|}\frac{\left\|Df(x')^{-1}(f(x) - f(x') - Df(x')(x - x'))\right\|}{\|x - x'\|} \\ &\leq 2K\frac{\|f(x) - f(x') - Df(x')(x - x')\|}{\|x - x'\|},\end{aligned}$$

which converges to zero by differentiability of f. This proves the claim $Df^{-1}(y) = [Df(x)]^{-1}$ where $y = f(x)$. ■

Remark 1.6: The inverse function theorem is *local*. In general we can't expect to find a single global inverse. For instance, let

$$f(x,y) := (e^y \cos(x), e^y \sin(x)).$$

One easily verifies

$$\det(Df(x,y)) = e^{-y} \neq 0.$$

However,

$$f(x+2k\pi, y) = f(x,y), \forall\, k \in \mathbb{Z},$$

so there is certainly no hope of a global inverse, for f is not even injective.

↪**Theorem 1.8** (Implicit Function Theorem): Let $F : \Omega \subset \mathbb{R}^n_x \times \mathbb{R}^m_y \to \mathbb{R}^m_y$ be a C^k map. Denote $X = (x,y) \in \mathbb{R}^n \times \mathbb{R}^m$, and let $X_0 = (x_0, y_0) \in \Omega$ with $F(X_0) = 0$. Writing $F = (F_1, \ldots, F_m)$, assume that

$$D_y F(X_0) = \begin{pmatrix} \frac{\partial F_1}{\partial y_1} & \cdots & \frac{\partial F_1}{\partial y_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial y_1} & \cdots & \frac{\partial F_m}{\partial y_m} \end{pmatrix}(X_0)$$

is invertible. Then, there exist neighborhoods U and V of $x_0 \in \mathbb{R}^n$ and $y_0 \in \mathbb{R}^m$ resp. and a unique C^k map $f : U \to V$ such that

$$F(x, f(x)) = 0, \quad \forall\, x \in U.$$

In other words, the level set of F is locally to x_0 the graph of some function f of the same regularity as F.

Proof. Define $G : \Omega \to \mathbb{R}^n \times \mathbb{R}^m$ by

$$G(x,y) := (x, F(x,y)).$$

Obviously G is C^k. We can apply the inverse function theorem to G near X_0; indeed,

$$DG(X_0) = \begin{pmatrix} I_{n\times n} & 0 \\ D_x F(X_0) & D_y F(X_0) \end{pmatrix},$$

which means

$$\det DG(X_0) = \det D_y F(X_0) \neq 0,$$

by assumption. Thus there exist neighborhoods W_1, W_2 of X_0, $(x_0, 0)$ respectively (since $(x_0, 0) = G(X_0)$) for which G^{-1} exists (and is C^k) from $W_2 \to W_1$. Then, there are neighborhoods $U \subset \mathbb{R}^n$ of x_0 and $V \subset \mathbb{R}^m$ of y_0 such that $U \times V \subset W_1$; set $Z = G(U \times V)$ (which is also open, with $Z \subset W_2$). Thus we can view

$$G : U \times V \to Z, \quad G^{-1} : Z \to U \times V,$$

which are both C^k maps. Since $G(x,y) = (x, F(x,y))$, we know that $G^{-1}(x,w) = (x, H(x,w))$ for all $(x,w) \in Z$. Here, $H : Z \to V$ is C^k since G is. Thus,

$$(x, F(x, H(x,w))) = G(x, H(x,w)) = (x,w),$$

using the definition of G in the first equality and the inverse fact in the second line. Thus, it follows that

$$F(x, H(x,w)) = w, \quad \forall\, (x,w) \in Z,$$

thus taking $f(x) := H(x,0)$ gives the proof. ■

↪**Corollary 1.2**: Let $F : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ be a $C^k(\Omega)$ function. Let $X = (x', y) \in \mathbb{R}^{n-1} \times \mathbb{R}$ and suppose $(x_0', y_0) \in \Omega$ with $\frac{\partial f}{\partial y}(x_0', y_0) \neq 0$. Then, there exist neighborhoods U and V of $x_0' \in \mathbb{R}^{n-1}$ and $y_0 \in \mathbb{R}$ and a unique $C^k(U)$ function $f : U \to V$ such that

$$\{F(x', y) = 0\} = \{y = f(x')\}, \quad (x', y) \in U \times V.$$

↪**Theorem 1.9** (Morse Lemma): Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ be a C^k function with $k \geq 3$. Let $0 \in \Omega$ be a critical point, i.e. $\nabla f(0) = 0$. Assume further $f(0) = 0$ and $\nabla^2 f(0)$ is invertible. There exist open sets U, V of $0 \in U \cap V$ and $g \in C^{k-2}(U)$, $g : U \to V$ with $g^{-1} : V \to U$, $g^{-1} \in C^2(V)$, such that

$$f(g(y)) = y_{\ell+1}^2 + \cdots + y_n^2 - \left(y_1^2 + \cdots + y_\ell^2\right),$$

for some $\ell \in \mathbb{Z} \cap [0, n]$.

§1.4 Taylor's Theorem in $\mathbb{R}^n$ and Lagrange Multipliers

Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$, $f \in C^{k+1}(\Omega)$. Let $x_0 \in \Omega$ and $|t|$ small. Consider

$$g(t) := f\left(x_0 + t\frac{x - x_0}{\|x - x_0\|}\right), \quad x \neq x_0, \quad g(0) = f(x_0).$$

Since $x_0 \in \Omega$ and Ω open, $x_0 + t\frac{x-x_0}{\|x-x_0\|} \in \Omega$ for t sufficiently small. By Taylor in 1-dimension,

$$g(t) = g(0) + g'(0)t + \frac{g''(0)t^2}{2!} + \cdots + \frac{g^{(k)}(0)t^k}{k!} + R_k(g)(t), \quad \frac{|R_k(g)(t)|}{|t|^k} \leq M\,|t| \text{ as } t \to 0.$$

Remark 1.7: This expansion suggests that one should be able to weaken the C^{k+1} assumption on g if we only require

$$\lim_{t \to 0} \frac{|R_k(g)(t)|}{|t|^k} = 0.$$

Indeed, we only really need $g \in C^{k-1}$ and $D\left(D^{k-1}\right)g(0)$ existing.

To get the Taylor expansion for $f(x)$ around x_0, we set $t = |x - x_0|$ and apply chain rule to $g(t)$. First, we compute $g^{(j)}(0)$; we get

$$g(0) = f(x_0),$$
$$g(\|x - x_0\|) = f(x).$$

By chain rule,

$$g'(0) = \sum_{j=1}^{n} \frac{\partial f}{\partial x_i}(x_0) \frac{x_j - x_j^0}{\|x - x_0\|} = \nabla f(x_0)^T \left(\frac{x - x_0}{\|x - x_0\|} \right).$$

Similarly,

$$\begin{aligned}
g''(t) &= \frac{\mathrm{d}}{\mathrm{d}t} g'(t) \\
&= \frac{\mathrm{d}}{\mathrm{d}t} \left(\nabla f \left(x_0 + t \frac{x - x_0}{\|x - x_0\|} \right)^T \frac{x - x_0}{\|x - x_0\|} \right) \\
&= \left(D^2 f \left(x_0 + t \frac{x - x_0}{\|x - x_0\|} \right) \frac{x - x_0}{\|x - x_0\|} \right) \cdot \frac{x - x_0}{\|x - x_0\|} \\
&= \left(\frac{x - x_0}{\|x - x_0\|} \right)^T D^2 f \left(x_0 + t \frac{x - x_0}{\|x - x_0\|} \right) \frac{x - x_0}{\|x - x_0\|} \\
&= \sum_{i,j=1}^{n} \frac{\partial^2 f}{\partial x_i \partial x_j} \left(x_0 + t \frac{x - x_0}{\|x - x_0\|} \right) \frac{\left(x_i - x_i^0\right)\left(x_j - x_j^0\right)}{\|x - x_0\|^2},
\end{aligned}$$

so that

$$g''(0) = \left[\frac{x - x_0}{\|x - x_0\|} \right]^T D^2 f(x_0) \cdot \left[\frac{x - x_0}{\|x - x_0\|} \right].$$

Similar computation can be used to compute $g^{(\ell)}$ for ℓ up to k. In general, we find that

$$g^{(\ell)}(0) = \sum_{i_1, \ldots, i_\ell = 1}^{n} \frac{\partial^\ell f}{\partial x_{i_1} \cdots \partial x_{i_\ell}}(x_0) \frac{x_{i_1} - x_{i_1}^0}{\|x - x_0\|} \cdots \frac{x_{i_\ell} - x_{i_\ell}^0}{\|x - x_0\|}.$$

Moreover, since $t^\ell = \|x - x_0\|^\ell$, the term in the Taylor expansion of g corresponding to $g^{(\ell)}(0)\frac{t^\ell}{\ell!}$ becomes $\frac{1}{\ell!}$ times the above, with the denominator $\|\ldots\|$ terms cancelling. This leads to the following theorem.

↪**Theorem 1.10** (Taylor in $\mathbb{R}^n$): Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}, f \in C^{k+1}(\Omega)$. Then, for any $x^0 \in \Omega$,

$$f(x) = f(x^0) + \sum_{j=1}^{k} \frac{1}{j!} \left[\sum_{i_1, \ldots, i_j = 1}^{n} \frac{\partial^j f}{\partial x_{i_1} \cdots \partial x_{i_j}} \left(x_{i_1} - x_{i_1}^0\right) \cdots \left(x_{i_\ell} - x_{i_j}^0\right) \right] + R_k(f)(x),$$

where

$$\frac{|R_k(f)(x)|}{\|x - x^0\|^k} \leq M_k \|x - x^0\| \quad \text{as } x \to x^0.$$

Remark 1.8: The k-th order Taylor polynomial of f is the quantity

$$P_k(f)(x) = f(x_0) + \sum_{j=1}^{k} \frac{1}{j!}\left[\sum_{i_1,\dots,i_j=1}^{n} \frac{\partial^j f}{\partial x_{i_1}\cdots\partial x_{i_j}}\left(x_{i_1} - x_{i_1}^0\right)\cdots\left(x_{i_\ell} - x_{i_j}^0\right)\right]$$

before the remainder, and is a "good approximation" to $f(x)$ for x near x^0, provided

$$\lim_{x\to x^0} \frac{|R_k(f)(x)|}{\|x - x^0\|^k} = 0.$$

If we just make this assumption on the remainder term, we get the following more general result with weaker assumptions:

↪**Theorem 1.11**: Let $f : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ be $C^{k-1}(\Omega)$ and assume $D^{k-1}f$ is differentiable at $x^0 \in \Omega$. Then the kth order Taylor expansion for f about x^0 from the previous theorem still holds, but with now

$$\lim_{x\to x^0} \frac{|R_k(f)(x)|}{\|x - x^0\|^k} = 0.$$

Remark 1.9: Of course, we lose the rate of decay of the remainder, but we gain fewer assumptions needed on f.

⊛ **Example 1.2**: Let $f(x,y) = e^x + \sin(xy)$. Then

$$f(0,0) = 1,$$

$$f_x(0,0) = 1, \quad f_y(0,0) = 0,$$

$$f_{xx}(0,0) = 1, \quad f_{xy}(0,0) = 1$$

$$f_{yx}(0,0) = 1, \quad f_{yy}(0,0) = 0.$$

Thus,

$$f(x,y) = 1 + x + \frac{x^2}{2} + xy + R_2(f)(x,y),$$

with

$$\left|\frac{R_s(f)(x,y)}{x^2+y^2}\right| \le M\sqrt{x^2+y^2}.$$

§1.5 Lagrange Multipliers

The basic problem is, given $f, g : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ both $C^1(\Omega)$, putting $\Sigma := \{x \in \Omega : g(x) = 0\}$, to find

$$\max f|_\Sigma, \quad \min f|_\Sigma.$$

We call $g : \Omega \to \mathbb{R}$ the "constraint function" (i.e., we are doing constrained optimization of f subject to g).

↪**Theorem 1.12**: Let f, g as above, and assume $Dg(x^0) \neq 0$ for all $x^0 \in \Sigma$. Then, if $f|_\Sigma$ has a max or min at $x_0 \in \Sigma$, then there exists $\lambda \in \mathbb{R}$ such that

$$\nabla f(x_0) = \lambda \nabla g(x_0).$$

Proof. Let $x^0 \in \Sigma$. Since by assumption $\nabla g(x_0) \neq 0$, then we can assume (by possibly relabelling the coordinates in $\mathbb{R}^n$) that $\frac{\partial g}{\partial x_n}(x^0) \neq 0$. We can write

$$\Sigma = \{g(x', x_n) = 0\},$$

with $x' = (x_1, \ldots, x_{n-1})$. By IFT, Σ then is of the form

$$x_n = h(x'),$$

for $x' \in \mathbb{R}^{n-1}$ for $(x', x_n) \in \Omega$ near $x^0 \in \Sigma$, where $h \in C^1$ locally to $x_0' := \left(x_1^0, \ldots, x_{n-1}^0\right)$. Now, we set

$$F = f|_\Sigma,$$

so that

$$F(x') = f(x', h(x'))$$

locally to x_0'. Since x' a local max/min of F, we know that $\nabla F(x') = 0$. Applying chain rule (valid since $h \in C^1$) above, we also know that

$$0 = \nabla_{x'} F(x_0') = \nabla_{x'} f(x_0', h(x_0')) = \left(\frac{\partial f}{\partial x_i}(x_0) + \frac{\partial h}{\partial x_i}(x_0') \frac{\partial f}{\partial x_n}(x_0) \right)_{i=1}^{n-1}. \quad (\dagger)$$

We also know that

$$0 = G(x') := g(x', h(x'))$$

for all x' local to x_0', and hence $\nabla G(x') = 0$ local to x_0' as well, i.e.

$$0 = \nabla_{x'} G(x_0') = \left(\frac{\partial g}{\partial x_i}(x_0) + \frac{\partial h}{\partial x_i'}(x_0') \frac{\partial g}{\partial x_n}(x_0) \right)_{i=1}^{n-1}. \quad (\ddagger)$$

Since $\frac{\partial g}{\partial x_n}(x_0) \neq 0$ by assumption, we use $(\ddagger)$ to isolate

$$\frac{\partial h}{\partial x_i'}(x_0') = -\left[\frac{\partial g}{\partial x_n}(x_0) \right]^{-1} \frac{\partial g}{\partial x_i}(x_0), \quad i = 1, \ldots, n-1.$$

Plugging these expressions into $(\dagger)$, we obtain

$$\frac{\partial f}{\partial x_i}(x_0) = \left[\frac{\partial g}{\partial x_n}(x_0) \right]^{-1} \frac{\partial f}{\partial x_n}(x_0) \frac{\partial g}{\partial x_i}(x_0), \quad i = 1, \ldots, n-1.$$

This implies

$$\nabla f(x_0) = \left(\frac{\partial f}{\partial x_1}(x_0), \dots, \frac{\partial f}{\partial x_{n-1}}(x_0), \frac{\partial f}{\partial x_n}(x_0)\right) = \underbrace{\left[\frac{\frac{\partial f}{\partial x_n}(x_0)}{\frac{\partial g}{\partial x_n}(x_0)}\right]}_{=\lambda}\left(\frac{\partial g}{\partial x_1}(x_0), \dots, \frac{\partial g}{\partial x_{n-1}}(x_0), \frac{\partial g}{\partial x_n}(x_0)\right) = \lambda \nabla g(x_0),$$

as claimed (remarking that equality for the first $n-1$ entries follows from our work above, and the last n is trivial since then the denominator of λ cancels with the nth entry of ∇g.) ■

↪**Theorem 1.13** (Max/min): Let $\Omega \subset \mathbb{R}^n$ be a domain with bounded closure, with $\Omega \subset \tilde{\Omega}$ and assume $f, g : \tilde{\Omega} \to \mathbb{R}$ are $C^1(\tilde{\Omega})$, where

$$\partial\Omega = \{x \in \tilde{\Omega} \mid g(x) = 0\}$$

(we say g the *defining function* of $\partial\Omega$). Then,

1. if $x^0 \in \partial\Omega$ a local max/min of $f|_{\partial\Omega}$ then $\nabla f(x^0) = \lambda \nabla g(x^0)$ for some $\lambda \in \mathbb{R}$,
2. if $x^0 \in \operatorname{int}(\Omega)$ a local max/min of f in $\operatorname{int}(\Omega)$, then $\nabla f(x^0) = 0$, and
3. the global max (resp. min) of $f|_{\overline{\Omega}}$ is largest (resp. smallest) value of f at all points in 1., 2.

Proof. Apply the previous result with $\Sigma = \partial\Omega$. ■

↪**Theorem 1.14** (Multiple Constraints): Let $f, g_i : \Omega \subset \mathbb{R}^n \to \mathbb{R}$ $i = 1, \dots, k$ are all $C^1(\Omega)$. Let

$$\Gamma_0 := \{x \in \Omega \mid g_i(x) = 0, \quad i = 1, \dots, k\}$$

and assume

$$\{\nabla g_1(x), \dots, \nabla g_k(x)\}$$

are linearly independent for $x \in \Gamma_0$. Then if $f|_{\Gamma_0}$ has max/min at $x^0 \in \Gamma_0$, there exist $\lambda_1, \dots, \lambda_k$ such that

$$\nabla f(x^0) = \sum_{i=1}^{k} \lambda_i \nabla g_i(x^0).$$

⊛ **Example 1.3**: Find the absolute max/min of $f(x_1, \dots, x_n) = x_1 + x_2$ in $D_1(0) = \{|x|^2 \leq 1\}$. We take $g(x) = |x|^2 - 1$. Since $\nabla f(x) = (1, 1, 0, \dots, 0)$ there cannot be any interior critical points. By Lagrange, at $x_0 \in \partial D_1$, a max/min occuring implies

$$\nabla f(x^0) = \lambda \nabla g(x^0) \Leftrightarrow (1, 1, 0, \dots, 0) = 2\lambda\left(x_1^0, \dots, x_n^0\right).$$

Note that λ can't be zero, thus $x_3^0, \dots, x_n^0 = 0$ and $x_1^0 = \frac{1}{2\lambda} = x_2^0$. Since $x_1^0, x_2^0 \in \partial D_1(0)$ as well, we know $\left(x_1^0\right)^2 + \left(x_2^0\right)^2 = 1$. This implies $\lambda = \pm\sqrt{\frac{1}{2}}$ and so $x_1^0 = x_2^0 = \pm\frac{1}{\sqrt{2}}$. We see that

$$f\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0, \dots, 0\right) = \sqrt{2}, \quad f\left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0, \dots, 0\right) = -\sqrt{2},$$

thus the global max of f is $\sqrt{2}$ and the global min of f is $-\sqrt{2}$.

§2 Riemann Integration

§2.1 Integration over Box Domains in $\mathbb{R}^n$

A *box* $B \subset \mathbb{R}^n$ is a closed and bounded subset of the form

$$B = [a_1, b_1] \times \cdots \times [a_n, b_n].$$

Given a function $f : B \to \mathbb{R}$, we want to define the Riemann integral (we will drop the "Riemann" specification from here on) of f over B, i.e. $\int_B f \, \mathrm{d}V$.

We know that in $n = 1$, for a function $h : [a, b] \to \mathbb{R}$ (assume nonnegative for simplicity), then

$$\int_a^b h(x) \, \mathrm{d}x \overset{\text{if it exists}}{=} \lim_{n \to \infty} \sum_{i=1}^{n} h(x_i) \Delta x_i,$$

where $a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b$ a partition of $[a, b]$ and $\Delta x_i = x_i - x_{i-1}$. With this as "motivation", say, we will define the integral of f over B.

↪**Definition 2.1** (Partition of a box): Let B a box as above. A *partition* of B is an ordered set

$$\mathcal{P} = \{P_i\}_{i=1}^n, \quad P_i = \left\{x_i^0, \ldots, x_i^{m_i}\right\},$$

where each P_i a partition of $[a_i, b_i]$ with $a_i = x_i^0 < x_i^1 < \cdots < x_i^{m_i - 1} < x_i^{m_i} = b_i$.

↪**Definition 2.2** (Refinement): Given two partitions $\mathcal{P} = \{P_i\}$, $\widetilde{\mathcal{P}} = \{\tilde{P}_i\}$, we say that $\widetilde{\mathcal{P}}$ is a *refinement* of $\mathcal{P}$ (and write $\widetilde{\mathcal{P}} \supset \mathcal{P}$) if for each $j = 1, \ldots, n$, $P_j \subset \tilde{P}_j$.

↪**Definition 2.3** (Subbox): Given a partition $\mathcal{P}$ of B, let

$$B_{i_1, i_2, \ldots, i_n} \subseteq B = \left[x_1^{i_1 - 1}, x_1^{i_1}\right] \times \cdots \times \left[x_n^{i_n - 1}, x_n^{i_n}\right]$$

be a *subbox* of B, for some $\{i_1, \ldots, i_n\}$ partition indices for $\mathcal{P}$ (i.e., $1 \leq i_j \leq m_j$ for $j = 1, \ldots, n$). We define then

$$\Delta x_j^{i_j} := x_j^{i_j} - x_j^{i_j - 1}, \quad j = 1, \ldots, n, 1 \leq i_j \leq m_j,$$

for the length of the jth interval in the subbox, and

$$\Delta V_{i_1, \ldots, i_n} := \Delta x_1^{i_1} \cdots \Delta x_n^{i_n}$$

as thus the volume of the subbox $B_{i_1, \ldots, i_n}$.

Heuristically, then, we'd like to define $\int_B f \, \mathrm{d}V$ as

$$\lim_{m_1 \to \infty} \cdots \lim_{m_n \to \infty} \sum_{i_1 = 1}^{m_1} \cdots \sum_{i_n = 1}^{m_n} f\left(x_1^{i_1}, \ldots, x_n^{i_n}\right) \Delta V_{i_1, \ldots, i_n}.$$

The key question, then, is when does the sum on the right-hand side exist?

↪**Definition 2.4** (Lower/Upper Riemann Sum): Let $f : B \to \mathbb{R}$ be bounded and $\mathcal{P}$ a partition of $\mathcal{B}$. Define the *lower Riemann sum* of f over B corresponding to partition $\mathcal{P}$ by

$$L(f, \mathcal{P}) := \sum_{i_1=1}^{m_1} \cdots \sum_{i_n=1}^{m_n} \left(\inf_{x \in B_{i_1,\ldots,i_n}} f(x) \right) \Delta V_{i_1,\ldots,i_n}.$$

Similarly define the *upper Riemann sum* (of f over B corresponding to partition $\mathcal{P}$) by

$$U(f, \mathcal{P}) := \sum_{i_1=1}^{m_1} \cdots \sum_{i_n=1}^{m_n} \left(\sup_{x \in B_{i_1,\ldots,i_n}} f(x) \right) \Delta V_{i_1,\ldots,i_n}.$$

Remark 2.1: This infs/sups are well-defined since f bounded.

↪**Lemma 2.1**: Let $f : B \to \mathbb{R}$ be bounded and $\widetilde{\mathcal{P}} \supset \mathcal{P}$ a refinement. Then,

$$L(f, \mathcal{P}) \leq L\left(f, \widetilde{\mathcal{P}}\right) \leq U\left(f, \widetilde{\mathcal{P}}\right) \leq U(f, \mathcal{P}).$$

Proof. The intermediate inequality is obvious.

Since $\widetilde{\mathcal{P}} \supset \mathcal{P}$, then there exists $N_{i_j} \in \mathbb{Z}^+$ and $i_{1,\ell_1}, \ldots, i_{n,\ell_n}$ (indices of the refined partition) such that

$$B_{i_1,\ldots,i_n} = \bigcup_{\ell_1=1}^{N_{i_1}} \cdots \bigcup_{\ell_n=1}^{N_{i_n}} \tilde{B}_{i_{1,\ell_1},\ldots,i_{n,\ell_n}},$$

with $\tilde{\cdot}$ used to denote objects belonging to the refinement. Thus,

$$\inf_{x \in B_{i_1,\ldots i_n}} f(x) \leq \inf_{x \in \tilde{B}_{i_{1,\ell_1},\ldots i_{n,\ell_n}}} f(x), \quad \sup_{x \in B_{i_1,\ldots i_n}} f(x) \geq \sup_{x \in \tilde{B}_{i_{1,\ell_1},\ldots i_{n,\ell_n}}} f(x).$$

Summing over the $i_1, \ldots, i_n$'s and multiplying by the appropriate volumes gives the result. ■

↪**Definition 2.5**: Let $f : B \to \mathbb{R}$ bounded. Define

$$\mathcal{U}(f) := \inf_{\mathcal{P}} \{U(f, \mathcal{P})\}, \quad \mathcal{L}(f) := \sup_{\mathcal{P}} \{L(f, \mathcal{P})\}.$$

Remark 2.2: If f bounded with $m \leq f \leq M$ over B, then

$$m \operatorname{vol}(B) \leq U(f, \mathcal{P}) \leq M \operatorname{vol}(B)$$

for any partition $\mathcal{P}$, similar for the lower Riemann sum. This implies these two quantities above are well-defined (infs/sups taken over bounded sets). Moreover, we take inf/sup respectively for the upper/lower sums because of the respective monotonicity properties we proved in the previous lemma.

↪**Definition 2.6** (Riemann integrable): Let $f : B \to \mathbb{R}$ be bounded. Then, we say that f is *Riemann integrable* (RI) over B if $\mathcal{L}(f) = \mathcal{U}(f)$, in which case we define

$$\int_B f \, \mathrm{d}V := \mathcal{L}(f) = \mathcal{U}(f).$$

↪**Lemma 2.2**: Let $f : B \to \mathbb{R}$ be bounded then f is RI over B iff $\forall\, \varepsilon > 0$, there exists a partition $\mathcal{P}$ of B such that

$$U(f, \mathcal{P}) - L(f, \mathcal{P}) \leq \varepsilon.$$

Proof. ($\Leftarrow$) We know $U(f, \mathcal{P}) \geq \mathcal{U}(f)$ and $L(f, \mathcal{P}) \leq \mathcal{L}(f)$, so this is immediate since then we conclude that for all $\varepsilon > 0$,

$$\mathcal{U}(f) \leq U(f, \mathcal{P}) \leq L(f, \mathcal{P}) + \varepsilon \leq \mathcal{L}(f) + \varepsilon.$$

Since this holds for all $\varepsilon > 0$, this implies $\mathcal{U}(f) - \mathcal{L}(f) \leq \varepsilon$ for all $\varepsilon > 0$ and thus must be equal, since $\mathcal{U}(f) \geq \mathcal{L}(f)$ always.

($\Rightarrow$) For any $\varepsilon > 0$, there exists a $\mathcal{P}_1$ with $L(f, \mathcal{P}_1) - \mathcal{L}(f) \leq \frac{\varepsilon}{2}$ and $\mathcal{P}_2$ with $\mathcal{U}(f) - U(f, \mathcal{P}_2) \leq \frac{\varepsilon}{2}$. Let $\mathcal{P} = \mathcal{P}_1 \cup \mathcal{P}_2$, then the result follows by monotonicity and triangle inequality. ■

↪**Theorem 2.1**: Let $f : B \to \mathbb{R}$ be in $C^0(B)$. Then f is Riemann integrable.

Proof. Since f continuous on a compact set it is uniformly continuous. Thus for any $\varepsilon > 0$, we can find a partition $\mathcal{P}$ such that for every subbox $B_{i_1,\ldots,i_n}$, we have

$$\sup_{x \in B_{i_1,\ldots,i_n}} f(x) - \inf_{x \in B_{i_1,\ldots,i_n}} f(x) < \frac{\varepsilon}{\mathrm{vol}(B)}.$$

This readily implies $U(f, \mathcal{P}) - L(f, \mathcal{P}) \leq \varepsilon$, so the result follows by the previous lemma. ■

↪**Lemma 2.3** (Mean Value Theorem for Integrals): Let $f : B \to \mathbb{R}$ be continuous. Then, there exists $x_0 \in B$ such that $\int_B f \, \mathrm{d}V = f(x_0)\, \mathrm{vol}(B)$.

Proof. Let $m := \inf_B |f|$, $M := \sup_B |f|$. Then

$$m \cdot \mathrm{vol}(B) \leq \int_B f \, \mathrm{d}V \leq M \cdot \mathrm{vol}(B).$$

Then we can just apply the intermediate value theorem to f. ■

§2.2 General Riemann Integration

We've defined the Riemann integral for f continuous over box domains, but we wish to define the integral for more general domains. Let Ω a bounded domain in $\mathbb{R}^n$ and let B a box with $\Omega \subset B$. Extend $f : \Omega \to \mathbb{R}$ to $F : B \to \mathbb{R}$ by

$$F(x) := \begin{cases} f(x) & x \in \Omega \\ 0 & x \in B \setminus \Omega \end{cases}.$$

Heuristically, then, we wish the integral to satisfy the following:

$$\int_B f \, \mathrm{d}V = \int_\Omega f \, \mathrm{d}V + \int_{B \setminus \Omega} F \, \mathrm{d}V = \int_\Omega f \, \mathrm{d}V.$$

↪**Definition 2.7**: Given $f : \Omega \to \mathbb{R}$ bounded, set

$$\int_\Omega f \, \mathrm{d}V := \int_B F \, \mathrm{d}V,$$

whenever the right-hand side exists.

This leaves the obvious next question - what happens when f not continuous? When is the integral defined?

↪**Definition 2.8**: Let $\Gamma \subset \mathbb{R}^n$ is of *content zero* if for all $\varepsilon > 0$, there exist closed box domains $B_1^0, \ldots, B_N^0 \subset \mathbb{R}^n$ such that $\Gamma \subset \bigcup_{n=1}^N B_i^0$ and $\sum_{i=1}^N \mathrm{vol}\left(B_i^0\right) < \varepsilon$.

Remark 2.3:

1. The closed boxes can be replaced with open boxes
2. If $D_1, \ldots, D_N$ are content zero so are their union

↪**Theorem 2.2**: Suppose $f : B \subset \mathbb{R}^n \to \mathbb{R}$ bounded. Let

$$\Gamma(f) := \{x \in B : f \text{ discontinuous at } x\}.$$

Then, if $\Gamma(f)$ has content zero, then $\int_B f \, \mathrm{d}V$ exists.

Proof. We prove using the lemma following the definition of Riemann integrability. Let $\varepsilon > 0$ and put $M := \sup_B |f|$. Since $\Gamma(f)$ has content zero, there exists $B_1^0, \ldots, B_N^0 \subset \mathbb{R}^n$ such that

$$\Gamma(f) \subset \bigcup_{i=1}^N B_i^0, \quad \text{and} \quad \sum_{i=1}^N \mathrm{vol}\left(B_i^0\right) < \frac{\varepsilon}{4(M+1)}.$$

Let P_0 be a partition of B such that each subrectangle $B_{i_1,\ldots,i_n}$ is either in some B_j^0, or doesn't intersect any such B_j^0 i.e. $B_{i_1,\ldots,i_n} \cap \left(\bigcup_{i=1}^N B_i^0\right) = \varnothing$. Let A be the union of these boxes that don't intersect $\bigcup_{i=1}^N B_i^0$. A is compact, being bounded ($A \subset B$) and closed (finite union of closed sets). Thus $f|_A$ is uniformly continuous, and so there exists a refined partition $\widetilde{\mathcal{P}}$ of $\mathcal{P}$ such that for each subrectangle $Q \in \widetilde{\mathcal{P}}$ that is a subset of A,

$$\sup_Q f - \inf_Q f < \frac{\varepsilon}{2(\mathrm{vol}(B)+1)}.$$

Then

$$\begin{aligned} U\left(f, \widetilde{\mathcal{P}}\right) - L\left(f, \widetilde{\mathcal{P}}\right) &= \sum_{Q \subset A} \left(\sup_Q f - \inf_Q f\right) \mathrm{vol}(Q) + \sum_{Q \in \widetilde{\mathcal{P}}, Q \not\subset A} \left(\sup_Q f - \inf_Q f\right) \mathrm{vol}(Q) \\ &\leq \frac{\varepsilon \, \mathrm{vol}(B)}{2(\mathrm{vol}(B)+1)} + (2M) \cdot \frac{\varepsilon}{4M+1} \leq \varepsilon. \end{aligned}$$

■

⊛ **Example 2.1**: Let $\Omega \subset \mathbb{R}^{n-1}$ compact and $g : \Omega \to \mathbb{R}$ continuous. Then $\Gamma := \{(x', g(x')) : x' \in \Omega\}$ is has content zero.

↪**Corollary 2.1**: Suppose $f : B \to \mathbb{R}$ bounded and $\Gamma(f)$ the union of finitely many graphs of continuous functions over closed, bounded domains in $\mathbb{R}^{n-1}$. Then f is integrable over B.

↪**Corollary 2.2**: Let $\Omega \subset \mathbb{R}^n$ closed and bounded and suppose $\partial\Omega$ the union of finitely many graphs of C^0 functions over closed domains in $\mathbb{R}^{n-1}$. Suppose $f : \Omega \to \mathbb{R}$ bounded and $\Gamma(f) \cap \Omega$ has content zero. Then $\int_\Omega f \, \mathrm{d}V$ exists.

There is a special case of the previous result that is of practical importance. Let $\Omega \subset \mathbb{R}^n$ closed and bounded and suppose $\Omega \subset \tilde{\Omega}$ a bounded a domain. Let $G \in C^1\left(\tilde{\Omega}\right)$ be a defining function for $\partial\Omega$, i.e.

$$\partial\Omega = \left\{G(x) = 0 : x \in \tilde{\Omega}\right\}, \quad \nabla G(x) \neq 0 \, \forall \, x \in \partial\Omega.$$

By inverse-function theorem, $\partial\Omega$ is the finitely many (since compact) graphs of C^1-functions over closed and bounded domains in $\mathbb{R}^{n-1}$. Thus, we have the following:

↪**Corollary 2.3**: Let $\Omega := \{G(x) \leq 0, x \in \mathbb{R}^n\}$ be bounded and assume $\nabla G(x) \neq 0$ whenever $G = 0$. Suppose $f : \Omega \to \mathbb{R}$ bounded and $\Gamma(f) \cap \Omega$ has content zero. Then $\int_\Omega f \, \mathrm{d}V$ exists.

§2.3 Basic Properties of Integrals

↪**Proposition 2.1**: Assume $\Omega \subset \mathbb{R}^n$ bounded and $f, g : \Omega \to \mathbb{R}$ are both Riemann integrable over Ω. Then,

1. $\int_\Omega (f + g) \, \mathrm{d}V = \int_\Omega f \, \mathrm{d}V + \int_\Omega g \, \mathrm{d}V$
2. $\int_\Omega \alpha f \, \mathrm{d}V = \alpha \int_\Omega f \, \mathrm{d}V, \quad \alpha \in \mathbb{R}$
3. $f \leq g \Rightarrow \int_\Omega f \, \mathrm{d}V \leq \int_\Omega g \, \mathrm{d}V$
4. If $\Omega = \Omega_1 \cup \Omega_2$ with each Ω_i bounded with $\Omega_1 \cap \Omega_2$ having content zero, then

$$\int_\Omega f \, \mathrm{d}V = \int_{\Omega_1} f \, \mathrm{d}V + \int_{\Omega_2} f \, \mathrm{d}V.$$

§2.4 Fubini's Theorem

Given $\Omega \subset \mathbb{R}^n$ bounded etc, how do we actually compute $\int_\Omega f \, \mathrm{d}V$? Consider $B = [a, b] \times [c, d] \subset \mathbb{R}^2$ (2 dimensions is sufficient to demonstrate the main results).

Given $f : B \to \mathbb{R}$ bounded, we define the iterated integrals

$$\int_a^b \int_c^d f(x, y) \, \mathrm{d}y \, \mathrm{d}x := \int_a^b \left[\int_c^d f(x, y) \, \mathrm{d}y \right] \mathrm{d}x,$$

similarly for the other order. We assume formally for now that both 1-dimensional integrals involved exists, i.e. that $\int_a^b f(x, y) \, \mathrm{d}x$ exists for $y \in [c, d]$ and $\int_c^d f(x, y) \, \mathrm{d}y$ exists for $x \in [a, b]$. Fubini essentially says that both of these iterated integrals are equal to $\int_B f \, \mathrm{d}V$.

Remark 2.4: For $\Omega \subset \mathbb{R}^2$ we often denote $\int_\Omega f \, \mathrm{d}V =: \iint_\Omega f \, \mathrm{d}A$.

↪**Theorem 2.3** (Fubini for Box Domains): Suppose $f : B \to \mathbb{R}$ is Riemann integrable with $B = [a,b] \times [c,d]$.

1. Assume $\int_c^d f(x,y) \, \mathrm{d}y$ exists for all $x \in [a,b]$. Then the iterated integral $\int_a^b \int_c^d f(x,y) \, \mathrm{d}y \, \mathrm{d}x$ exists and equals $\int_\Omega f \, \mathrm{d}V$.
2. The same statement, but with the other iterated integral.

Proof. For any $\varepsilon > 0$ choose $\mathcal{P}$ of B such that $U(f,\mathcal{P}) - L(f,\mathcal{P}) < \varepsilon$. This partition induces partitions

$$\mathcal{P}_x = \{x_0 = a < x_1 < \cdots < x_{n-1} < x_n = b\}, \quad \mathcal{P}_y = \{y_0 = c < y_1 < \cdots < y_{m-1} < y_m = d\}.$$

Set $F(x) = \int_c^d f(x,y) \, \mathrm{d}y = \sum_{j=1}^m \int_{y_{j-1}}^{y_j} f(x,y) \, \mathrm{d}y$. We compute

$$U(F,\mathcal{P}_x) = \sum_{i=1}^n \left(\sup_{x \in [x_{i-1},x_i]} F(x) \right) \Delta x_i, \quad L(F,\mathcal{P}_x) = \sum_{i=1}^n \left(\inf_{x \in [x_{i-1},x_i]} F(x) \right) \Delta x_i.$$

Thus, we find

$$\begin{aligned}
U(F,\mathcal{P}_x) &= \sum_{i=1}^n \left(\sup_{x \in [x_{i-1},x_i]} \sum_{j=1}^m \int_{y_{j-1}}^{y_j} f(x,y) \, \mathrm{d}y \right) \Delta x_i \\
&\leq \sum_{i=1}^n \sum_{j=1}^m \left[\sup_{x \in [x_{i-1},x_i]} \int_{y_{j-1}}^{y_j} f(x,y) \, \mathrm{d}y \right] \Delta x_i \\
&\leq \sum_{i=1}^n \sum_{j=1}^m \left[\sup_{x \in [x_{i-1},x_i]} \sup_{y \in [y_{j-1},y_j]} f(x,y) \right] \Delta y_j \Delta x_i \\
&= \sum_{i=1}^n \sum_{j=1}^m \left[\sup_{(x,y) \in B_{ij}} f(x,y) \right] \Delta B_{ij}, \quad B_{ij} := [x_{i-1},x_i] \times [y_{j-1},y_j] \\
&= U(f,\mathcal{P}).
\end{aligned}$$

Similar work shows $L(F,\mathcal{P}) \geq L(f,\mathcal{P})$ and thus

$$U(F,\mathcal{P}) - L(F,\mathcal{P}) \leq U(f,\mathcal{P}) - L(f,\mathcal{P}) < \varepsilon$$

which gives the proof. ■

2.4.1 Fubini for More General Domains

↪**Definition 2.9**: A domain $\Omega \subset \mathbb{R}^2$ is called *y-simple* if it is of the form

$$\Omega = \{a \leq x \leq b, \varphi_1(x) \leq y \leq \varphi_2(x)\},$$

for some $\varphi_1, \varphi_2 \in C^0([a,b])$. Similarly it is called *x-simple* if

$$\Omega = \{c \leq y \leq d, \psi_1(y) \leq x \leq \psi_2(y)\}$$

for some $\psi_1, \psi_2 \in C^0([c,d])$. Finally, it is called *simple* if it is both x- and y-simple, and *elementary* if it is x- or y-simple.

↪**Theorem 2.4** (Fubini): Suppose $\Omega \subset \mathbb{R}^2$ is bounded and f integrable over Ω.

1. If Ω is y-simple and for all $x \in [a,b]$, $\int_{\varphi_1(x)}^{\varphi_2(x)} f(x,y)\,\mathrm{d}y$ exists, then
$$\int_\Omega f\,\mathrm{d}V = \int_a^b \int_{\varphi_1(x)}^{\varphi_2(x)} f(x,y)\,\mathrm{d}y\,\mathrm{d}x.$$
2. The analogous statement for x-simple.
3. If both x- and y-simple, we can compute the Riemann integral with either iterated integral in 1., 2..

Proof. Suppose Ω y-simple. Choose a box domain $B = [a,b] \times [c,d]$ for which $\Omega \subset B$. Then
$$\iint_\Omega f\,\mathrm{d}A = \iint_B F\,\mathrm{d}A, \quad F := \begin{cases} f & \text{in } \Omega \\ 0 & \text{else} \end{cases}.$$
Note F RI since f is. By Fubini on box domains,
$$\iint_B F\,\mathrm{d}A = \int_a^b \int_c^d F(x,y)\,\mathrm{d}y\,\mathrm{d}x = \int_a^b \int_{\varphi_1(x)}^{\varphi_2(x)} F(x,y)\,\mathrm{d}y\,\mathrm{d}x = \int_a^b \int_{\varphi_1(x)}^{\varphi_2(x)} f(x,y)\,\mathrm{d}y\,\mathrm{d}x,$$
since $F(x,\cdot)$ is zero outside $y \in [\varphi_1(x), \varphi_2(x)]$. The proof for x-simple domains follows similarly. ■

⊛ **Example 2.2**: Compute $\iint_\Omega x\,\mathrm{d}A$ where $\Omega = \{(x,y) : 0 \le x \le \sqrt{\pi}, 0 \le y \le \sin(x^2)\}$. It's clear $(x,y) \mapsto x$ continuous on Ω and since Ω is bounded and y-simple, the double integral exists and moreover can be computed using Fubini's. We have
$$\begin{aligned}\iint_\Omega x\,\mathrm{d}A &= \int_0^{\sqrt{\pi}} \int_0^{\sin(x^2)} x\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_0^{\sqrt{\pi}} x\sin(x^2)\,\mathrm{d}x \quad [u = x^2] \\ &= \frac{1}{2}\int_0^{\pi} \sin(u)\,\mathrm{d}u = 1.\end{aligned}$$

⊛ **Example 2.3**: Compute $I = \int_0^1 \int_y^1 e^{x^2}\,\mathrm{d}x\,\mathrm{d}y$. We can't compute the inner integral in closed-form. But, if we let $\Omega := \{(x,y) : 0 \le y \le 1, y \le x \le 1\}$, we find that $I = \iint_\Omega e^{x^2}\,\mathrm{d}A$. We can rewrite this domain as $\Omega = \{(x,y) : 0 \le x \le 1, 0 \le y \le x\}$, and thus by Fubini's (everything that needs to be continuous is continuous),
$$\begin{aligned} I &= \int_0^1 \int_0^x e^{x^2}\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_0^1 xe^{x^2}\,\mathrm{d}x \\ &= \frac{1}{2}(e-1).\end{aligned}$$

2.4.2 Fubini in $\mathbb{R}^n$

Given a box domain $B \subset \mathbb{R}^n$, write

$$\underline{\int_B} f \, \mathrm{d}V := \mathcal{L}(f), \quad \overline{\int_B} f \, \mathrm{d}V := \mathcal{U}(f)$$

(here, f needn't be RI).

↪**Theorem 2.5**: Let $B = B_1 \times B_2$ be a box domain in $\mathbb{R}^n$ where $B_1 \subset \mathbb{R}^k$, $B_2 \subset \mathbb{R}^{n-k}$ are also boxes. Suppose $f : B \to \mathbb{R}$ is Riemann integrable. Set

$$\ell(x) := \underline{\int_{B_2}} f(x,y) \, \mathrm{d}V(y), \quad u(x) := \overline{\int_{B_2}} f(x,y) \, \mathrm{d}V(y), \quad x \in B_1$$

and

$$\tilde{\ell}(y) := \underline{\int_{B_1}} f(x,y) \, \mathrm{d}V(x), \quad \tilde{u}(y) := \overline{\int_{B_1}} f(x,y) \, \mathrm{d}V(x), \quad y \in B_2.$$

Then, $\ell(x)$ and $u(x)$ are Riemann integrable over B_1 and $\tilde{\ell}(y)$ and $\tilde{u}(y)$ are Riemann integrable over B_2. Moreover,

$$\int_B f \, \mathrm{d}V = \int_{B_1} \ell(x) \, \mathrm{d}V(x) = \int_{B_1} u(x) \, \mathrm{d}V(x),$$
$$\int_B f \, \mathrm{d}V = \int_{B_2} \tilde{\ell}(y) \, \mathrm{d}V(y) = \int_{B_2} \tilde{u}(y) \, \mathrm{d}V(y).$$

Proof. The proof follows in the same way as in $\mathbb{R}^2$ by noting the following:

$$L(f, \mathcal{P}) \leq L(\ell(x), \mathcal{P}_x), \quad U(u(x), \mathcal{P}_x) \leq U(f, \mathcal{P}).$$

This implies, suping/infing over all $\mathcal{P}$,

$$\mathcal{L}(f) \leq \mathcal{L}(\ell(x)), \quad \mathcal{U}(u(x)) \leq \mathcal{U}(f)$$

and thus since f Riemann integrable,

$$0 = \mathcal{U}(f) - \mathcal{L}(f) \geq \mathcal{U}(u(x)) - \mathcal{L}(\ell(x)) \geq 0 \Rightarrow \mathcal{U}(u(x)) = \mathcal{L}(\ell(x))$$

thus $\ell(x)$ Riemann integrable and in particular $\mathcal{L}(f) = \mathcal{L}(\ell(x)) = \mathcal{U}(\ell(x)) = \mathcal{U}(f)$. Similar holds in the y-variables. ■

Remark 2.5: If for all $x \in B_1, f(x,y)$ is integrable in B_2 (i.e. $\int_{B_2} f(x,y) \, \mathrm{d}V(y)$ exists), then

$$\int_B f \, \mathrm{d}V = \int_{B_1} \left[\int_{B_2} f(x,y) \, \mathrm{d}V(y) \right] \mathrm{d}V(x).$$

Similarly, if $\int_{B_1} f(x,y) \, \mathrm{d}V(x)$ exists for all $y \in B_2$, then

$$\int_B f \, \mathrm{d}V = \int_{B_2} \left[\int_{B_1} f(x,y) \, \mathrm{d}V(x) \right] \mathrm{d}V(y).$$

2.4.3 Fubini's in $\mathbb{R}^3$

↪**Definition 2.10**: A bounded domain $W \subset \mathbb{R}^3$ is called *z-simple* if there exists an elementary region $\Omega \subset \mathbb{R}^2$ and $\eta_1, \eta_2 \in C^0(\Omega)$ such that

$$W = \{(x,y,z) : (x,y) \in \Omega, \eta_1(x,y) \leq z \leq \eta_2(x,y)\}.$$

Analogous definitions hold for x-, y-simple as in the $\mathbb{R}^2$ case. We additionally say W elementary if it is simple in some axis.

↪**Theorem 2.6** (Fubini in $\mathbb{R}^3$): Suppose $W \subset \mathbb{R}^3$ is bounded and z-simple and $f : W \to \mathbb{R}$ is RI. Assume that for all $(x,y) \in \Omega$, $\int_{\eta_1(x,y)}^{\eta_2(x,y)} f(x,y,z)\,\mathrm{d}z$ exists. Then, it is Riemann integrable over Ω and

$$\iiint_W f\,\mathrm{d}V = \int_\Omega \left[\int_{\eta_1(x,y)}^{\eta_2(x,y)} f(x,y,z)\,\mathrm{d}z\right] \mathrm{d}V(x,y).$$

Similar holds for x-, y- simple domains.

Proof. Follows as in $\mathbb{R}^2$. ■

§2.5 Change of Variables Formula

Let $\Omega \subset \mathbb{R}^n$ be a bounded domain. We defined $\int_\Omega f\,\mathrm{d}V$ by looking at $L(f,\mathcal{P})$ and $U(f,\mathcal{P})$ where $\mathcal{P}$ was a *rectangular partition*. Can we use other partitions of Ω that are *not* rectangular?

↪**Definition 2.11**: Let $\Omega \subset \mathbb{R}^n$ be a domain with content zero boundary. A *regular partition* $\mathcal{P}_{\text{reg}} = \{D_k\}_{k=1}^N$ is a collection of subdomains of $D_k \subset \Omega$ such that
1. $\Omega = \bigcup_{k=1}^N D_k$ and $\text{Content}(\partial D_k) = 0$ for all $k = 1, \ldots, N$, and
2. $\text{Content}(D_k \cap D_\ell) = 0$ for $k \neq \ell$.

↪**Definition 2.12**: $f : \Omega \to \mathbb{R}$ is bounded where Ω a bounded domain and $\mathcal{P}_{\text{reg}} = \{D_k\}_{k=1}^N$ a regular partition of Ω. Define

$$U(f,\mathcal{P}_{\text{reg}}) := \sum_{k=1}^N \left(\sup_{D_k} f\right) \text{vol}(D_k), \quad L(f,\mathcal{P}_{\text{reg}}) := \sum_{k=1}^N \left(\inf_{D_k} f\right) \text{vol}(D_k).$$

↪**Lemma 2.4**: Let $f : B \to \mathbb{R}$. Then f is Riemann integrable over B iff for any $\varepsilon > 0$ there exists a regular partition $\mathcal{P}_{\text{reg}}$ of B such that

$$U(f,\mathcal{P}_{\text{reg}}) - L(f,\mathcal{P}_{\text{reg}}) < \varepsilon.$$

Proof. ($\Rightarrow$) is immediate, since a rectangular partition is regular and we already know the analogous statement for such partitions.

($\Leftarrow$) Fix $\varepsilon > 0$ and let $\mathcal{P}_{\text{reg}} = \{D_k\}$ is such that

$$U(f,\mathcal{P}_{\text{reg}}) - L(f,\mathcal{P}_{\text{reg}}) < \varepsilon.$$

Then, $\Gamma := \bigcup_k \partial D_k$ has content zero by assumption, and so there exist boxes $B_1, \ldots, B_m$ such that $\Gamma \subset \bigcup_j B_j$ and $\sum_j \operatorname{vol}\left(B_j\right) < \frac{\varepsilon}{2(M+1)}$ where $M := \sup_B |f|$. Pick a box partition $\mathcal{P}$ such that

$$\sum_{\substack{B_i \in \mathcal{P} \\ \mathcal{B}_i \cap \Gamma \neq \varnothing}} \operatorname{vol}(B_i) < \frac{\varepsilon}{2(M+1)},$$

and for each $B_j \subset \mathcal{P}$ with $B_j \cap \Gamma = \varnothing$ we have that $B_j \subset D_\ell$ for some ℓ (obtained by refining). Note that if B_j is such that $B_j \cap \Gamma$,

$$0 \leq \sup_{B_j} f - \inf_{B_j} f \leq \sup_{D_\ell} f - \inf_{D_\ell} f.$$

Then

$$\begin{aligned}
U(f, \mathcal{P}) - L(f, \mathcal{P}) &= \sum_{\substack{B_i \in \mathcal{P} \\ B_i \cap \Gamma \neq \varnothing}} \left(\sup_{B_i} f - \inf_{B_i} f \right) \operatorname{vol}(B_i) + \sum_{\substack{B_i \in \mathcal{P} \\ B_i \cap \Gamma = \varnothing}} \left(\sup_{B_i} f - \inf_{B_i} f \right) \operatorname{vol}(B_i) \\
&\leq 2M \sum_{\substack{B_i \in \mathcal{P} \\ B_i \cap \Gamma \neq \varnothing}} \operatorname{vol}(B_i) + \sum_{D_\ell} \left(\sup_{D_\ell} f - \inf_{D_\ell} \right) \operatorname{vol}(D_\ell) \\
&< (2M) \frac{\varepsilon}{2(M+1)} + U\left(f, \mathcal{P}_{\text{reg}}\right) - L\left(f, \mathcal{P}_{\text{reg}}\right) \\
&< 2\varepsilon.
\end{aligned}$$

■

↪**Theorem 2.7** (Change of Variables): Let $\Omega, \Omega' \subset \mathbb{R}^n$ bounded domains and let $T : \Omega' \to \Omega$ be a C^1 map that is bijective from Ω' onto Ω. Then, for any $f : \Omega \to \mathbb{R}$ which is Riemann-integrable over Ω, then

$$\int_\Omega f \, \mathrm{d}V = \int_{\Omega'} (f \circ T) |\det DT| \, \mathrm{d}V.$$

To prove, we need to first observe how the volume of a box B changes under application of T. Let $B' \subset \Omega'$ be a small rectangular box. Since T C^1, by Taylor expansion, we have for any $y_0 \in B' \subset \mathbb{R}^n$,

$$T(y) = T(y_0) + DT(y_0)(y - y_0) + E_1(y, y_0)$$

where $\lim_{y \to y_0} \frac{\|E_1(y, y_0)\|}{\|y - y_0\|} = 0$. Since $DT(y_0)$ is linear,

$$\operatorname{vol}(T(B')) = |\det(DT(y_0))| \operatorname{vol}(B') + E_2(B')$$

where for any $\varepsilon > 0$, there exists $\delta > 0$ such that $\|E_1(y, y_0)\| \leq \varepsilon \|y - y_0\|$ whenever $0 < \|y - y_0\| < \delta$. Suppose $B' = \left\{ y \in \mathbb{R} : \max_{j=1}^n \left| y_j - y_j^0 \right| \leq \delta \right\}$ (a box of side length δ). Using the fact that $\|y - y_0\| \leq \sqrt{n} \max_{j=1}^n \left| y_j - y_j^0 \right|$. Then,

$$E_2(B') \leq (\varepsilon \|y - y_0\|)^n \leq \left(\varepsilon \sqrt{n} \max_j \left| y_j - y_j^0 \right| \right)^n \leq (\sqrt{n})^n \varepsilon^n \left(\max_j \left| y_j - y_j^0 \right| \right)^n = \left(\sqrt{n}^n \varepsilon^n \right) \operatorname{vol}(B') \leq C_n \varepsilon \operatorname{vol}(B'),$$

where C_n some dimensional constant. Thus,

$$\frac{|E_2(B')|}{\operatorname{vol}(B')} \to 0 \text{ as } \|y - y_0\| \to 0.$$

Remark that since $T \in C^1(B')$, $|\det(DT)| \in C^0(B')$ and so uniformly continuous in B', so given $\varepsilon > 0$, we can find $\delta > 0$ such that $|E_2(B')| \leq \varepsilon \operatorname{vol}(B')$ and $\|\det(DT(y_0))| - |\det DT(y)\| \leq \varepsilon$ for any $y \in B'$ with $\|y - y_0\| < \delta$ (?) and thus, for all $\varepsilon > 0$, there exists a B' of sidelength $\leq \delta$,

$$\operatorname{vol}(TB') = |\det DT(y)| \operatorname{vol}(B') + \tilde{E}(B'), \quad |\tilde{E}(B')| \leq \varepsilon \operatorname{vol}(B')$$

Proof. We use the observations above and the previous lemma on regular partitions. For any $\varepsilon > 0$, we can find $\delta > 0$ and a box B' with sidelength $< \delta$ such that

$$\operatorname{vol}(TB') = |\det DT(y)| \operatorname{vol}(B') + E_2(B'),$$

for any $y \in B'$, where $|E_2(B')| \leq \varepsilon \operatorname{vol}(B')$.

Choose a box partition $\mathcal{P}' = \{B'_j\}$ on Ω'. Since $T : \Omega' \to \Omega$ a C^1 bijection onto Ω, $\mathcal{P}_{\text{reg}} = \{D_i = T(B'_i)\}$ is a regular partition on Ω (why?). We assume that $U(f, \mathcal{P}_{\text{reg}}) - L(f, \mathcal{P}_{\text{reg}}) < \varepsilon$ and $|E_2(B'_i)| \leq \varepsilon \operatorname{vol}(B'_i)$ for all $B'_i \in \mathcal{P}'$, by refinement and since f integrable. Then,

$$U(f, \mathcal{P}_{\text{reg}}) \geq U((f \circ T)|\det(DT)|, \mathcal{P}') - 2M\varepsilon \operatorname{vol}(\Omega'),$$

and similarly

$$L(f, \mathcal{P}_{\text{reg}}) \leq L((f \circ T)|\det(DT)|, \mathcal{P}') + 2M\varepsilon \operatorname{vol}(\Omega')$$

where $M = \sup_\Omega |f|$. This implies

$$\begin{aligned}\varepsilon > U(f, \mathcal{P}_{\text{reg}}) - L(f, \mathcal{P}_{\text{reg}}) &\geq U((f \circ T)|\det(DT)|, \mathcal{P}') - L((f \circ T)|\det(DT)|, \mathcal{P}') - 4M\varepsilon \operatorname{vol}(\Omega') \\ &\Rightarrow U((f \circ T)|\det(DT)|, \mathcal{P}') - L((f \circ T)|\det(DT)|, \mathcal{P}') < \varepsilon(1 + 4M \operatorname{vol}(\Omega')).\end{aligned}$$

Since ε arbitrary, this proves the result. ■

⊛ **Example 2.4** (Polar Coordinates): Take $x = r\cos\theta, y = r\sin\theta$, so $r = \sqrt{x^2 + y^2}$. Let $T(r, \theta) = (x, y)$ from $\Omega' \to \Omega$ with $\Omega' = \{(r, \theta) \in (0, a] \times [0, 2\pi)\}$ and $\Omega = \{(x, y) : x^2 + y^2 \leq a^2\}$. Then

$$\iint_{\{x^2+y^2 \leq a^2\}} f(x, y)\, \mathrm{d}A = \iint_{\Omega'} f(r\cos(\theta), r\sin(\theta))|\det(DT)|\, \mathrm{d}r\, \mathrm{d}\theta,$$

where

$$\det(DT) = \det\left(\begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}\right) = r.$$

By Fubini, then, we get

$$\iint_{\{x^2+y^2 \leq a^2\}} f(x, y)\, \mathrm{d}A = \int_0^{2\pi} \int_0^a f(r\cos(\theta), r\sin(\theta)) r\, \mathrm{d}r\, \mathrm{d}\theta.$$

⊛ **Example 2.5** (Spherical Coordinates): Let $x = \rho \sin\varphi \cos\theta, y = \rho \sin\varphi \sin\theta, z = \rho\cos\varphi$ and $T(\rho,\varphi,\theta) = (x,y,z)$ with $\theta \in [0,2\pi), \varphi \in [0,\pi)$ and $(0,a]$. In this case

$$|\det(DT)| = |\det\begin{pmatrix} \sin\varphi\cos\theta & \rho\cos\varphi\cos\theta & -\rho\sin\varphi\sin\theta \\ \sin\varphi\sin\theta & \rho\cos\varphi\sin\theta & \rho\sin\varphi\cos\theta \\ \cos\varphi & -\sin\varphi & 0 \end{pmatrix}| = \rho^2\sin\varphi$$

so that

$$\iiint_{\{x^2+y^2+z^2\le a^2\}} f(x,y,z)\,\mathrm{d}V = \int_0^{2\pi}\int_0^{\pi}\int_0^{a} f(\rho\sin\varphi\cos\theta, \rho\sin\varphi\sin\theta, \rho\cos\varphi).$$

⊛ **Example 2.6**: Compute $\iint_\Omega (y-x)\cos(xy)\,\mathrm{d}A$, with Ω the region bounded by $x+y=1, x+y=4, xy=-1$ and $xy=-4$ and $y-x\ge 0$. Let $u = x+y$, $v = xy$ and $T(u,v) = (x,y)$ with $T : \Omega' \to \Omega$, where we now have $\Omega' = \{(u,v) \in [1,4]\times[-4,-1]\}$. Moreover, we have

$$(x+y)^2 - (x-y)^2 = 4xy \Rightarrow u^2 - (x-y)^2 = 4v \Rightarrow |x-y| = \sqrt{u^2-4v}.$$

Finally,

$$|\det(DT^{-1})| = |\det\left(\begin{pmatrix} 1 & 1 \\ y & x \end{pmatrix}\right)| = |x-y| \Rightarrow |\det DT| = \frac{1}{|x-y|}.$$

This implies

$$\iint_\Omega (x-y)\cos(xy)\,\mathrm{d}A = \int_1^4\int_{-4}^{-1}\cos(v)\,\mathrm{d}v\,\mathrm{d}u = 3[\sin(-1)-\sin(-4)].$$

§3 Vector Fields, Curves and Surfaces

§3.1 Integration along Curves

↪**Definition 3.1**: A curve in $\mathbb{R}^n$ is a continuous function $\gamma : I \to \mathbb{R}^n$ where $I \subset \mathbb{R}$ some interval.

↪**Definition 3.2**: Let γ be a C^1 curve defined on $I = [0,L]$. Define the integral over γ of a real-valued function as

$$\int_\gamma f\,\mathrm{d}s := \int_0^L f(\gamma(t))|\gamma'(t)|\,\mathrm{d}t.$$

§3.2 Integration of Vector Fields

↪**Definition 3.3**: A *vector field* on $\mathbb{R}^n$ is a function $F : \Omega \subset \mathbb{R}^n \to \mathbb{R}^n$. We usually assume F C^0 on Ω.

How can we integrate a vector field along a path γ? (think: the work done by F along γ)

Let $\gamma : [a,b] \to \mathbb{R}^n$ be a C^1 path, and consider

$$\Delta s_i := \gamma(t_i + \Delta t_i) - \gamma(t_i)$$

for Δt_i *small*. This is a vector.

Thus, since γ is C^1,

$$\Delta s_i = \gamma'(t_i)\Delta t_i + R(t, \Delta t_i),$$

where $\frac{\|R(t,\Delta t_i)\|}{\Delta t_i} \to 0$ as Δt_i. We get:

$$F(\gamma(t)) \cdot \Delta s_i = F(\gamma(t_i))[\gamma'(t_i)\Delta t_i + R(t_i, \Delta t_i)].$$

Takeing $\mathcal{P} = \{t_i\}_{i=1}^N$ of $[a, b]$, we can repeat this over every partition. This motivates:

↪**Definition 3.4**: Define the *total work done* by F along γ by

$$\int_\gamma (F) \cdot (\mathrm{d}s) := \lim_{N\to\infty} \sum_{i=1}^N (F)(\gamma(t_i))(\Delta s_i),$$

as defined above. This is equal to

$$\int_a^b (F)(\gamma(t)) \cdot \gamma'(t)\,\mathrm{d}t,$$

assuming the Riemann integral exists, where γ lives on $[a, b]$.

↪**Definition 3.5**: Let γ a piecewise smooth path and F a vector field defined in a neighborhood of γ. The *line integral* (also called *path integral*) of F along γ is

$$\int_\gamma F \cdot \mathrm{d}s = \int_c^b F(\gamma(t)) \cdot \gamma'(t)\,\mathrm{d}t.$$

If γ is regular, i.e. $\gamma'(t) \neq 0$, let $T(t) = \frac{\gamma'(t)}{\|\gamma'(t)\|}$ be the *unit tangent field*. Then, we see that

$$\int_\gamma F \cdot \mathrm{d}s = \int_\gamma F(\gamma(t)) \cdot T(t),$$

where $\mathrm{d}s \equiv \|\gamma'(t)\|\,\mathrm{d}t$ the "arclength element".

Formally, we write $\operatorname{vec} \mathrm{d}s = \operatorname{vec} T\,\mathrm{d}s$.

↪**Lemma 3.1**: If γ a piecewise-smooth path and $\tilde{\gamma}$ a reparametrization of γ, then if F a C^0 vector field, then

$$\int_\gamma F \cdot \mathrm{d}s = \varepsilon \int_{\tilde{\gamma}} F \cdot \mathrm{d}s,$$

where $\varepsilon = +1$ if an orientation-preserving reparametrization, and $\varepsilon = -1$ if an orientation-reversing reparametrization.

3.2.1 Conservative Vector Fields

↪**Definition 3.6**: A vector field $F : \Omega \to \mathbb{R}^n$ is *conservative* in Ω if there exists a C^1 function $f : \Omega \to \mathbb{R}$ such that $F(x) = \nabla f(x)$ for all $x \in \Omega$.

↪**Theorem 3.1**: Suppose F a conservative vector field in a neighborhood of a piecewise-smooth path γ. Then

$$\int_\gamma F \cdot \mathrm{d}s = f(\gamma(b)) - f(\gamma(a)).$$

↪**Corollary 3.1**: If γ closed, then $\int_\gamma F \cdot \mathrm{d}s = 0$.

Proof. (Of Theorem)

$$\begin{aligned}\int_\gamma F \cdot \mathrm{d}s &= \int_a^b \nabla f(\gamma(t)) \cdot \gamma'(t)\,\mathrm{d}t \\ &= \int_a^b \frac{\mathrm{d}}{\mathrm{d}t} f(\gamma(t))\,\mathrm{d}t \\ &= f(\gamma(b)) - f(\gamma(a)).\end{aligned}$$

■

If $C = \cup_i C_i$ a finite union of curves C_i, we extend our definition to

$$\int_C F \cdot \mathrm{d}s = \sum_i \int_{C_i} F \cdot \mathrm{d}s.$$

3.2.2 Poincaré Lemma

The question we aim to address here is, when is $F : \Omega \to \mathbb{R}^n$ conservative?

A necessary condition is as follows.

↪**Theorem 3.2**: Let $F = (F_1, \ldots, F_n)$ a C^1 conservative vector field. Then,

$$\frac{\partial F_i}{\partial x_j}(x) = \frac{\partial F_j}{\partial x_i}(x)$$

for all i, j and $x \in \Omega$.

Proof. We have $F = \nabla f$ for $f \in C^1(\Omega)$; then $\frac{\partial F_i}{\partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$, so the claim follows by Clairault's. ■

When is this condition sufficient?

§3.3 Poincaré Lemma

↪**Lemma 3.2**: Suppose $g(x, y)$ is C^1 in $\Omega \times A$ where $\Omega \subset \mathbb{R}^n$ open and $A \subset \mathbb{R}^m$ is compact. Then,

$$\frac{\partial}{\partial x_i}\left(\int_A g(x,y)\,\mathrm{d}y\right) = \int_A \frac{\partial}{\partial x_i} g(x,y)\,\mathrm{d}y$$

for any $i \in [n]$ and $x \in \Omega$.

Proof. Fix $r > 0$ so that $B_r(x) \subset \Omega$. Let $h \in \mathbb{R}$ with $|h| < r$. By the mean-value theorem,

$$g(x+he_i,y)-g(x,y)=\frac{\partial g}{\partial x_i}\left(x^h,y\right)\cdot h,\quad x^h\in B_r(x).$$

Since $\overline{B_r(x)}\times A$ is compact, $\frac{\partial g}{\partial x_i}$ is uniformly continuous on this set. Thus for any $\varepsilon>0$, there exists $\delta>0$ such that

$$\left|\partial_{x_i}g\left(x^h,y\right)-\partial_{x_i}g(x,y)\right|<\varepsilon,\quad \forall\,|h|<\delta, y\in A.$$

Then,

$$\begin{aligned}\left|\overbrace{\frac{\int_A g(x+he_i,y)\,\mathrm{d}y-\int_A g(x,y)\,\mathrm{d}y}{h}}^{\text{difference quotient for integral}}-\int_A \partial_{x_i}g(x,y)\,\mathrm{d}y\right| &\leq \int_A\left|\frac{\partial g}{\partial x_i}\left(x^h,y\right)-\frac{\partial g}{\partial x_i}(x,y)\right|\mathrm{d}y\\ &\leq \varepsilon\,\mathrm{vol}(A),\end{aligned}$$

for all $\varepsilon>0$ with $|h|<\delta$. Since ε arbitrary, this proves the claim. ■

This tells us when we can differentiate under an integral sign.

↪**Definition 3.7** (Star-Shaped Region): $\Omega\subset\mathbb{R}^n$ *star-shaped* if there exists $p_0\in\Omega$, which we call a *center*, such that the line segment $\ell(p_0,x)$ connecting p_0 to x is contained entirely within Ω for all $x\in\Omega$.

↪**Theorem 3.3** (Poincaré Lemma): Suppose $\Omega\subset\mathbb{R}^n$ is star-shaped and F is a $C^1(\Omega,\mathbb{R}^n)$ vector field. Then, F is conservative iff

$$\frac{\partial F_i}{\partial x_j}=\frac{\partial F_j}{\partial x_i}$$

for all $i,j=1,\ldots,n$.

Remark 3.1: The statement holds for more general, so-called "contractible domains", but we limit our exposition here.

Proof. We know the "$\Rightarrow$" direction already. Conversely, suppose wlog that $p_0=0$ is a "center" of our star-shaped domain. The idea to construct our f such that $F=\nabla f$ is to integrate F along the straight-line curve connecting x to 0 for each x. Formally, define $\gamma_x(t)=tx, 0\leq t\leq 1$, and define

$$f(x)=\int_{\gamma_x}F\cdot\mathrm{d}s=\int_0^1 F(tx)\cdot x\,\mathrm{d}t=\sum_{i=1}^n\int_0^1 F_i(tx)x_i\,\mathrm{d}t.$$

We need to compute the gradient of this function. Since $(t,x)\mapsto F_i(tx)x_i$ clearly C^1 and $[0,1]$ compact, we can apply the lemma and differentiate in x under the integral sign, and we find

$$
\begin{aligned}
\frac{\partial f}{\partial x_j}(x) &= \int_0^1 \sum_{i=1}^n \left[\frac{\partial F_i}{\partial x_j}(tx) t x_i + F_i(tx)\delta_{i,j} \right] \mathrm{d}t \\
&= \int_0^1 F_j(tx) + \sum_{i=1}^n t x_i \frac{\partial F_i}{\partial x_j}(tx)\, \mathrm{d}t \\
\text{(integrate by parts on first summand)} \quad &= \int_0^1 \sum_{i=1}^n t x_i \frac{\partial F_i}{\partial x_j}(tx)\, \mathrm{d}t + \left[t F_j(tx) \right]_0^1 - \int_0^1 t \sum_{i=1}^n \frac{\partial F_j}{\partial x_i} x_i(tx)\, \mathrm{d}t \\
&= \int_0^1 \sum_{i=1}^n t x_i \underbrace{\left[\frac{\partial F_i}{\partial x_j}(tx) - \frac{\partial F_j}{\partial x_i}(tx) \right]}_{=0 \text{ by assumption}} \mathrm{d}t + F_j(x) = F_j(x),
\end{aligned}
$$

as claimed. ■

§3.4 Integration on Surfaces

↪**Definition 3.8**: A parametrized surface $S \subset \mathbb{R}^3$ is the image of a map $\Phi : D \to S$,

$$S = \{(x(u,v), y(u,v), z(u,v)) = \Phi(u,v) : (u,v) \in D\},$$

where $D \subset \mathbb{R}^2$ some bounded domain. If Φ is C^k we call S a C^k surface. Note that

$$\Phi_u = (\partial_u x, \partial_u y, \partial_u z), \quad \Phi_v = (\partial_v x, \partial_v y, \partial_v z)$$

are tangent to S.

↪**Definition 3.9**: For a regular C^1 parametrization $\Phi : D \to S$, define

$$\operatorname{area}(S) := \iint_D \|\Phi_u \times \Phi_v\| \, \mathrm{d}A,$$

when it exists.

↪**Definition 3.10**: Let $S = \Phi(D)$ a C^1 parametrized surface and suppose $f \in C^0(S)$. Then we define

$$\iint_S f \, \mathrm{d}S := \iint_D f(\Phi) \|\Phi_u \times \Phi_v\| \, \mathrm{d}A.$$

↪**Proposition 3.1**: In the special case that S is the graph of a C^1 function $g : D \to \mathbb{R}$ (i.e. $S = \{(x,y,g(x,y)) : (x,y) \in D\}$), then this integral becomes

$$\iint_D f(x,y,g(x,y)) \sqrt{1 + g_x^2 + g_y^2} \, \mathrm{d}A.$$

Proof. This follows from computing

$$\Phi_x = (1,0,g_x), \quad \Phi_y = \left(0,1,g_y\right) \Rightarrow \Phi_x \times \Phi_y = \left(-g_x, -g_y, 1\right).$$

■

↪**Definition 3.11**: Given $S = \Phi(D)$ a C^1 parametrized surface, we say S is regular if the normal $\Phi_u \times \Phi_v$ never vanishes over D, i.e. if Φ_u, Φ_v are never linearly dependent. If S regular, we can choose

$$N = \pm \frac{\Phi_u \times \Phi_v}{\|\Phi_u \times \Phi_v\|}$$

to be the *unit normal*; there are two choices of sign, leading to different *orientations* of S.

↪**Definition 3.12**: Let F a C^0 vector field on a regular parametrized surface $S = \Phi(D)$. Then, we define

$$\iint_S F \cdot \mathrm{d}s := \iint_D F \cdot N \|\Phi_u \times \Phi_v\| \,\mathrm{d}u \,\mathrm{d}v = \iint_D (F \circ \Phi) \cdot (\Phi_u \times \Phi_v) \,\mathrm{d}A.$$

Note that this is defined up to orientation of S; however, it isn't clear from the definition that this is independent of parametrization.

↪**Definition 3.13**: Given $\Phi : D \to S, \widetilde{\Phi} : \tilde{D} \to S$ regular C^1 parametrizations of the same surface, we say $\widetilde{\Phi}$ is a reparametrization of Φ if there exists a bijective C^1 map $h : \tilde{D} \to D$ with C^1 inverse, such that $\widetilde{\Phi}(s,t) = (\Phi \circ h)(s,t)$.

We say $\widetilde{\Phi}$ *orientation-preserving* if

$$\frac{(\Phi_u \times \Phi_v)(h(s,t))}{\|(\Phi_u \times \Phi_v)(h(s,t))\|} = \frac{\left(\widetilde{\Phi}_s \times \widetilde{\Phi}_t\right)(s,t)}{\left\|\left(\widetilde{\Phi}_s \times \widetilde{\Phi}_t\right)(s,t)\right\|},$$

and *reversing* if the same equality holds with a negative in front of the second term.

↪**Lemma 3.3**: Let $\widetilde{\Phi}, \Phi$ as above with $h : \tilde{D} \to D$ written as $h(s,t) = (u(s,t), v(s,t)) \in C^1(\tilde{D}, D)$ a change of parameters. Then h is orientation-preserving iff $\det\left(\frac{\partial(u,v)}{\partial(s,t)}\right) > 0$, and reversing iff $\det\left(\frac{\partial(u,v)}{\partial(s,t)}\right) < 0$, where $\frac{\partial(u,v)}{\partial(s,t)}$ the Jacobian matrix of h.

Proof. Follows from the chain rule upon computing $\widetilde{\Phi}_s$ and $\widetilde{\Phi}_t$, namely we find

$$\widetilde{\Phi}_s \times \widetilde{\Phi}_t = \det\left(\frac{\partial(u,v)}{\partial(s,t)}\right)(\Phi_u \times \Phi_v),$$

from which the claim readily follows. ■

From this lemma, it readily follows that the previous definition of the integral of a vector field over a surface is well-defined (modulo orientation) regardless of choice of parametrization, by applying the chain of variables formula and the previous lemma. We'll often denote a positively/negatively oriented surface by S^+, S^- depending on the choice of sign of the normal vector. As an immediate corollary, we find

$$\iint_{S^\pm} F \cdot \mathrm{d}s = -\iint_{S^\mp} F \cdot \mathrm{d}s.$$

§3.5 Green's Theorem and its Consequences

↪**Definition 3.14**: A surface $S \subset \mathbb{R}^3$ is said to be a *regular surface with C^1 boundary* ∂S if S is compact and if for all $p \in S$, there exists $r, \delta > 0$ with $B_r(\rho) \subset \mathbb{R}^3$ and $D = \{-\delta < u < \delta, -\delta < v < \delta\} \subset \mathbb{R}^2$ and a regular parametrization map $\Phi : D \to \mathbb{R}^3$ such that either

1. $S \cap B_r(\rho) = \Phi(D)$, or
2. $S \subset B_r(\rho) = \Phi(D^+)$ where $D_+ := \{-\delta < u < \delta, 0 \leq v < \delta\}$. Points in 1. are called *interior charts* of S and points in 2. are called *boundary charts of S*; the boundary of S is defined as $\partial S = \{\text{all boundary points of } S\}$. The local parametrized surfaces are called coordinate charts.

↪**Definition 3.15**: A regular surface $S \subset \mathbb{R}^3$ is orientable if the unit normal $n \in C^0(S, \mathbb{R}^3)$.

In general there are two choices of unit normal at any point of a surface of $\mathbb{R}^3$ ("in/out"). Given a regular orientable surface $S \subset \mathbb{R}^3$, we say it is positively oriented if:

1. If $\partial S = \varnothing$, the unit normal is pointing "outwards" towards the "unbounded" region of the ambient $\mathbb{R}^3$
2. If $\partial S \neq \varnothing$, then we choose a "positive/counterclockwise" orientation for the curve ∂S which induces an orientation on S by the right-hand rule.

We are now in the position to state and prove Green's, which relates line integrals of vector fields over ∂D to double integrals over a planar domain D. We say that a bounded planar domain $D \subset \mathbb{R}^2$ is positive if the unit normal (with respect to orientation on boundary) is $\lambda = (0, 0, 1)$ (equivalently, we can embed D into $\mathbb{R}^3$ as a (planar) surface, then positive orientation is the same as that defined above for surfaces).

↪**Theorem 3.4** (Green): Let $D \subset \mathbb{R}^2$ be a positively oriented bounded piecewise C^1 domain and $F = (P, Q) : D \to \mathbb{R}^2$ a vector field on D. Then

$$\int_{\partial D} F \cdot \mathrm{d}s = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathrm{d}A.$$

Proof. On ∂D, we have the short-hand $\mathrm{d}s = i\,\mathrm{d}x + j\,\mathrm{d}y = (\mathrm{d}x, \mathrm{d}y)$ where $i = (1, 0), j = (0, 1)$. We can write

$$\begin{aligned} \int_{\partial D} F \cdot \mathrm{d}s &= \int_{\partial D} (P, Q) \cdot (\mathrm{d}x, \mathrm{d}y) \\ &= \int_{\partial D} P\,\mathrm{d}x + Q\,\mathrm{d}y. \end{aligned}$$

Assume first D is a simple domain

$$D = \{a \leq x \leq b, \varphi_1(x) \leq y \leq \varphi_2(x)\} = \{c \leq y \leq d, \psi_1(y) \leq x \leq \psi_2(y)\}$$

with boundary C. We claim $-\iint_D \frac{\partial P}{\partial y}\,\mathrm{d}A = \int_C P\,\mathrm{d}x$. Write

$$
\begin{aligned}
C_1 &= \{a \le x \le b, y = \varphi_1(x)\} \\
C_2 &= \{a \le x \le b, y = \varphi_2(x)\} \\
B_1 &= \{x = a, \varphi_1(a) \le y \le \varphi_2(a)\} \\
B_2 &= \{x = b, \varphi_1(b) \le y \le \varphi_2(b)\}.
\end{aligned}
$$

C_1^+, C_2^+ are oriented $a \mapsto b$ and B_1^+, B_2^+ are oriented from $\varphi_1(a)$ to $\varphi_2(a), \varphi_1(b)$ to $\varphi_2(b)$ respectively. By Fubini,

$$
\begin{aligned}
\iint_D \frac{\partial P}{\partial y}\,\mathrm{d}A &= \int_a^b \int_{\varphi_1(x)}^{\varphi_2(x)} \frac{\partial P}{\partial y}\,\mathrm{d}y\,\mathrm{d}x \\
\text{(FTC)} \quad &= \int_a^b (P(x,\varphi_2(x)) - P(x,\varphi_1(x)))\,\mathrm{d}x \\
&= \int_{C_2^+} P\,\mathrm{d}x - \int_{C_1^+} P\,\mathrm{d}x = \int_{C_2^+ \cup C_{\bar{1}}} P\,\mathrm{d}x \\
&= -\int_{C_{\bar{2}} \cup C_1^+} P\,\mathrm{d}x + \int_{B_1^+} P\,\mathrm{d}x - \int_{B_2^+} P\,\mathrm{d}x \\
&= -\int_{C_{\bar{2}} \cup C_1^+ \cup (B_2^+) \cup B_{\bar{1}}} P\,\mathrm{d}x \\
&= -\int_C P\,\mathrm{d}x,
\end{aligned}
$$

where we use the fact that

$$
\int_{B_1^+} P\,\mathrm{d}x - \int_{B_2^+} P\,\mathrm{d}x = 0
$$

since x is constant on B_1 and B_2 to add them in to our computations.

Similarly, one shows that

$$
\iint_D \frac{\partial Q}{\partial x}\,\mathrm{d}A = \int_{\partial D} Q\,\mathrm{d}y.
$$

Combining these identities gives the proof.

For a more general domain, one case use the implicit function theorem and the piecewise-C^1 assumption to reduce the domain to a union of simple regions, which sum up to give the general result. ■

↪**Corollary 3.2**:

$$
\int_{\partial D} x\,\mathrm{d}y = -\int_{\partial D} y\,\mathrm{d}x = \frac{1}{2}\int_{\partial D} x\,\mathrm{d}y - y\,\mathrm{d}x = \operatorname{area}(D),
$$

where we use as in the previous theorem the notation, where if $\gamma = (\gamma_1, \gamma_2) : I \to \mathbb{R}^2$ is a parametrization of ∂D, then

$$
\mathrm{d}x = \gamma_1'(t)\,\mathrm{d}t,
$$

etc (and the input is evaluated on $(x, y) = \gamma(t)$).

Proof. Set $F = (0, x)$. By Green's,

$$\int_{\partial D} F \cdot \mathrm{d}s = \int_D \mathrm{d}A = \mathrm{area}(D).$$

On the other hand, the left-hand side is readily seen to be equal to $\int_{\partial D} x \, \mathrm{d}y$. Similar proofs with $F = (y, 0)$, $F = (-y, x)$ yield the other results. ■

Exercise 3.1: Compute $\mathrm{area}(D)$ via Green's, where $D = \left\{ \frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1 \right\}$.

Solution. The boundary is given parametrically in two parts, $x \in [-a, a]$ and $y = \pm b\sqrt{1 - x^2/a^2}$. The plus and minus of y give the same so by the previous corollary,

$$\mathrm{area}(D) = 2b \int_{-a}^{a} \sqrt{1 - x^2/a^2} \, \mathrm{d}x = \pi ab,$$

where we compute the final integral via a trig substitution. ■

↪**Corollary 3.3**: Let $D = D_1 \setminus D_2$ where $\partial D_1, \partial D_2$ are piecewise C^1 with orientations induced from $D_1 + D_2$. If $F = (P, Q)$ a C^1 vector field with $\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}$ on D, then

$$\int_{\partial D_1} F \cdot \mathrm{d}s = \int_{\partial D_2} F \cdot \mathrm{d}s.$$

Proof. With the given orientations, $\partial D = \partial D_1 - \partial D_2$. By Green's, $\int_{\partial D} F \cdot \mathrm{d}s = 0$ since $P_x = Q_y$. On the other hand, by linearity and given the orientations, $\int_{\partial D} F \cdot \mathrm{d}s = \int_{\partial D_1} F \cdot \mathrm{d}s - \int_{\partial D_2} F \cdot \mathrm{d}s$. Combined, these give the answer. ■

↪**Theorem 3.5**: With the same assumptions as in Green's,

$$\int_{\partial D} F \cdot n \, \mathrm{d}s = \iint_D \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \, \mathrm{d}A$$

where n the inward normal vector to ∂D.

Proof. Suppose ∂D parametrized by $(x(t), y(t))$ for $t \in [a, b]$, so that $n = \frac{-\dot{y}, \dot{x}}{\sqrt{\dot{x}^2 + \dot{y}^2}}$.

$$\begin{aligned}
\int_{\partial D} F \cdot n \, \mathrm{d}s &= \int_a^b (P, Q)(x(t), y(t)) \cdot n(t) \sqrt{x'(t)^2 + y'(t)^2} \, \mathrm{d}t \\
&= \int_a^b (P(x(t), y(t)), Q(x(t), y(t))) \cdot (y'(t), x(t)) \, \mathrm{d}t \\
&= \int_a^b P(x(t), y(t)) y'(t) \, \mathrm{d}t - \int_a^b Q(x(t), y(t)) x'(t) \, \mathrm{d}t \\
&= \int_{\partial D} \tilde{F} \, \mathrm{d}s, \quad \tilde{F} := (-Q, P) \\
\text{Green's} \quad &= \iint_D \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \, \mathrm{d}A.
\end{aligned}$$

At the penultimate line, one can alternatively arrive at the conclusion by the abuse of notation

$$\int_a^b P(x(t), y(t)) y'(t) \, \mathrm{d}t = \int_{\partial D} P \, \mathrm{d}x = \iint_D \frac{\partial P}{\partial x} \, \mathrm{d}A$$

also by Green's. ■

3.5.1 Harmonic Functions

Let $\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$ be the Laplacian operator in two variables. We call a function $u \in C^2(\Omega)$ *harmonic* provided $\Delta u \equiv 0$ in Ω.

↪**Corollary 3.4**: Let u be a harmonic function in a domain $D \subset \mathbb{R}^2$ with piecewise C^1 domain. Then $\int_{\partial D} \nabla u \cdot n \, \mathrm{d}s = 0$.

Proof. Take $F = \nabla u$ in the previous corollary. ■

↪**Corollary 3.5** (Green's Identities): Let $\varphi, \psi \in C^2(D)$, where $D \subset \mathbb{R}^2$ with piecewise C^1 domain with positive orientation. Then the following hold:

1. $\iint_D \varphi \Delta \psi \, \mathrm{d}A = -\iint_D \nabla \varphi \cdot \nabla \psi \, \mathrm{d}A + \int_{\partial D} \varphi \nabla \psi \cdot n \, \mathrm{d}s$
2. $\iint_D [\varphi \Delta \psi - \psi \Delta \varphi] \, \mathrm{d}A = \int_{\partial D} \varphi \nabla \psi \cdot n - \psi \nabla \varphi \cdot n \, \mathrm{d}s$

Proof.

1. follows from the previous-previous corollary applied to $F = \varphi \nabla \psi$.
2. follows from 1. by swapping the roles of φ, ψ and subtracting the following identities from each other.

■

§3.6 Stoke's Theorem

↪**Definition 3.16**: Let $F = (F_1, F_2, F_3) : \mathbb{R}^3 \to \mathbb{R}^3$ a C^1-vector field. Define

$$\begin{aligned} \operatorname{curl} F &:= \det \begin{pmatrix} e_1 & e_2 & e_3 \\ \partial_x & \partial_y & \partial_z \\ F_1 & F_2 & F_3 \end{pmatrix} \\ &:= \left(\partial_{x_2} F_3 - \partial_{x_3} F_2, \partial_{x_3} F_1 - \partial_{x_1} F_3, \partial_{x_1} F_2 - \partial_{x_2} F_1 \right). \end{aligned}$$

↪**Theorem 3.6** (Stoke's): Let S be a compact, oriented surface with positive orientation and piecewise C^1 boundary ∂S. Let $F : S \to \mathbb{R}^3$ a C^1 vector field. Then

$$\iint_S \operatorname{curl} F \cdot \mathrm{d}S = \int_{\partial S} F \cdot \mathrm{d}s.$$

Proof. Recall that for a vector field G, $\int_S G \cdot \mathrm{d}S := \int_S G \cdot n \, \mathrm{d}S$ where n the (in this case, outward) unit normal to S.

Step 1: assume first S is C^2 and ∂S is piecewise C^1. Since S regular, for all $p \in \partial S$, S is locally given by a graph over a a coordinate plane by the implicit function theorem. For instance if, suppose, S is locally given by $z = g(x, y)$, then $\Phi(x, y) = (x, y, g(x, y)) \in S$ gives a local parametrization of S, and induces positive unit normal

$$n = \frac{-g_x, -g_y, 1}{\sqrt{1 + g_x^2 + g_y^2}}.$$

We can always pick g such that $(0,0,g(0,0)) = p$. By compactness, we can decompose S into a finite union of "subsurfaces"

$$S = \bigcup_{\ell=1}^{N} S_\ell.$$

such that $S_\ell^{\text{int}} \cap S_k^{\text{int}} = \varnothing$ for all $k \neq \ell$, and such that each S_ℓ is locally a graph over a coordinate plane for each $\ell = 1, \ldots, N$. Then, we note that

$$\int_{\partial S} F \cdot \mathrm{d}s = \sum_{\ell=1}^{N} \int_{\partial S_\ell} F \cdot \mathrm{d}s, \quad \iint_S \operatorname{curl} F \cdot \mathrm{d}S = \sum_{\ell=1}^{N} \iint_{S_\ell} \operatorname{curl} F \cdot \mathrm{d}S,$$

since the overlapping boundary regions will cancel each other out by orientation. Thus, it suffices to prove Stoke's in the case that S is a graph. By rotation, translating, and rescaling, we may assume that S is a C^2 graph over $D \subset \mathbb{R}^2_{x,y}$ with $D = [-1,1] \times [0,1]$. Then, on the one hand,

$$\int_S \operatorname{curl} F \cdot \mathrm{d}S = \iint_D \big(\partial_y F_3 - \partial_z F_2\big)(-g_x) + (\partial_z F_1 - \partial_x F_3)\big(-g_y\big) + \partial_x F_2 - \partial_y F_1 \, \mathrm{d}A.$$

We now consider the integral over ∂S. We see first that $\partial S = \Phi(\partial D) = \{(x,y,g(x,y)) : (x,y) \in \partial D\}$. Thus, by chain rule, $\mathrm{d}s = \big(\mathrm{d}x, \mathrm{d}y, g_x \, \mathrm{d}x + \mathrm{d}g_y \, \mathrm{d}y\big)$ so

$$\int_{\partial S} F \cdot \mathrm{d}s = \int_{\partial D} (F_1 + F_3 g_x) \, \mathrm{d}x + \int_{\partial D} \big(F_2 + F_3 g_y\big) \, \mathrm{d}y = \int_{\partial D} G \cdot \mathrm{d}s,$$

where

$$G(x,y) := \big(F_1 + F_3 g_x, F_2 + F_3 g_y\big),$$

where, crucially, each F_i is evaluated at $(x,y,g(x,y))$. Applying Green's to this integral and applying a lot of chain and product rule gives the result; one will expect higher-order derivatives of g to appear, but these will (should!) cancel.

Step 2: Suppose now $S \in C^1$ with pw-C^1 boundaries. The heuristic idea is to "approximate" S with C^2 surfaces and show that this approximation respects the identity. We won't show the details. ■

↪**Corollary 3.6**: Suppose S_1, S_2 are two oriented surfaces with $\partial S_1 = \partial S_2 = \gamma$, and suppose S_1 and S_2 induce opposite orientations on γ. Then, for any $F \in C^1$,

$$\iint_{S_1} \operatorname{curl} F \cdot \mathrm{d}S = -\iint_{S_2} \operatorname{curl} F \cdot \mathrm{d}S.$$

↪**Corollary 3.7**: With S as above, suppose $\partial S = \varnothing$ and $F \in C^1(S)$. Then $\iint_S \operatorname{curl} F \cdot \mathrm{d}S = 0$.

↪**Corollary 3.8**: Suppose $F = \nabla f$ for some $f \in C^2$ and S is an oriented surface with $\partial S = \gamma_1 - \gamma_2$. Then

$$\int_{\gamma_1} F \cdot \mathrm{d}s = \int_{\gamma_2} F \cdot \mathrm{d}s.$$

Proof. One checks that $\operatorname{curl}(\nabla f) = 0$. ■

§3.7 Divergence Theorem

↪**Theorem 3.7**: Let $W \subset \mathbb{R}^3$ be a bounded, solid region in $\mathbb{R}^3$ with a piecewise smooth boundary $S = \partial W$ with positive orientation, and F a C^1 vector field on $W \to \mathbb{R}^3$. Then,

$$\iiint_W \operatorname{div} F \, \mathrm{d}V = \iint_S F \cdot \mathrm{d}S,$$

where if $F = (F_1, F_2, F_3)$, $\operatorname{div}(F) = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$.

↪**Corollary 3.9**: Under the same assumptions as above, if $f, g \in C^2(W)$, then

$$\iiint_W [g\Delta f - f\Delta g] \, \mathrm{d}V = \iint_{\partial W} [g\nabla f \cdot \mathrm{d}S - f\nabla g \cdot \mathrm{d}S].$$

Proof. As in Stoke's theorem, it suffices to prove for W a simple region. Clearly, it suffices to prove that

$$\iiint_W \frac{\partial F_3}{\partial z} \, \mathrm{d}V = \iint_S F_3 \hat{k} \cdot \mathrm{d}S,$$

where $\hat{k} = (0, 0, 1)$, by linearity. Assume

$$W = \{(x, y, z) : (x, y) \in D, g(x, y) \leq z \leq h(x, y)\}.$$

Fubini's gives

$$\begin{aligned}\iiint_W \frac{\partial F_3}{\partial z} \, \mathrm{d}V &= \iint_D \int_{g(x,y)}^{h(x,y)} \frac{\partial F_3}{\partial z} \, \mathrm{d}z \, \mathrm{d}x \, \mathrm{d}y \\ &= \iint_D [F_3(x, y, h(x, y)) - F_3(x, y, g(x, y))] \, \mathrm{d}x \, \mathrm{d}y.\end{aligned}$$

We can decompose the boundary $\partial W = S$ into three parts,

$$\begin{aligned}S_1 &= \{(x, y, g(x, y)) : (x, y) \in D\} \Rightarrow n = -\frac{-g_x, -g_y, 1}{\sqrt{1 + g_x^2 + g_y^2}} \\ S_2 &= \{(x, y, h(x, y)) : (x, y) \in D\} \Rightarrow n = \frac{-h_x, -h_y, 1}{\sqrt{1 + h_x^2 + h_y^2}} \\ S_3 &= \{\langle n, k \rangle = 0\}.\end{aligned}$$

The relevant integral over S_3 is zero. Over S_1, we get

$$\iint_{S_1} F_3(x, y, h(x, y)) - F_3(x, y, g(x, y)) \, \mathrm{d}x \, \mathrm{d}y,$$

since $\hat{k} \cdot n = 1$ in each case. ■

↪**Corollary 3.10**: Suppose $W, W_1, \ldots, W_k \subset \mathbb{R}^n$ are regions as in the previous theorem, with

$$W_i \cap W_j = \varnothing, i \neq j, \quad W_j \subset W^{\text{int}}, \quad p_j \in W_j^{\text{int}} \; \forall \, j = 1, \ldots, k$$

for some points $p_1, \ldots, p_k$. If F a C^1 vector field on $W - \{p_1, \ldots, p_k\}$ and div $F = 0$ for all $x \in W - \{p_1, \ldots, p_k\}$. Then,

$$\iint_{\partial W} F \cdot \mathrm{d}S = \sum_{j=1}^{k} \iint_{\partial W_j} F \cdot \mathrm{d}S.$$

Proof. Apply the divergence theorem to $W - \bigcup_{j=1}^{k} W_j$. ■

Exercise 3.2 (Gauss's Law): Let $r = (x, y, z)$. Then

$$\iint_{\partial W} \frac{r}{\|r\|^3} \, \mathrm{d}S = \begin{cases} 4\pi & 0 \in W \\ 0 & \text{else} \end{cases}$$

§4 Practice Questions

Question 4.1: *Let $f(x,y) = \frac{x^4y^2}{x^2+y^2}$ when $(x,y) \neq 0$ and $f(0,0) = 0$. Prove that f is differentiable everywhere in $\mathbb{R}^2$.*

Solution. If $(x,y) \neq (0,0)$, one can take partial derivatives and find that they are both continuous, thus f differentiable away from $(0,0)$. At $(0,0)$, we claim $Df(0,0) = 0$. For $h = (h_1, h_2) \in \mathbb{R}^2$, we find

$$\frac{|f(h_1,h_2) - f(0,0)|}{\|h\|} = \underbrace{\frac{h_1^4}{\left(h_1^2+h_2^2\right)^{1/2}}}_{\leq h_1^3} \underbrace{\frac{h_2^2}{h_1^2+h_2^2}}_{\leq 1} \leq h_1^3 \underset{\|h\|\to 0}{\to} 0.$$

■

Question 4.2: *Given the curve $C = \{y^2 + xy + \sin x = 1\}$, prove that C can be written as $y = g(x)$ for some C^1 function g near $\left(\frac{\pi}{2}, 0\right)$. Is the same true near $(0,1)$?*

Solution. Let $G(x,y) = y^2 + xy + \sin x - 1$. This is clearly C^1. We compute

$$\nabla G(x,y) = (y + \cos x, 2y + x).$$

At $\left(\frac{\pi}{2}, 0\right)$, we see $\nabla G\left(\frac{\pi}{2}, 0\right) = \left(0, \frac{\pi}{2}\right)$ ■

Question 4.3: *Set $f(x,y) = \frac{\sin(xy)}{xy}$ when $xy \neq 0$ and $f(0,y) = f(x,0) = 0$. Prove that f is Riemann integrable on $B := [-1,1] \times [-1,1]$.*

Solution. Its clear that $f(x,y)$ is continuous away from the line segments $L_x = \{(x,0) : -1 \leq x \leq 1\}, L_y = \{(0,y) : -1 \leq y \leq 1\} \subset B$. Thus, if we can show content$\left(L_x \cup L_y\right) = 0$, we'll be done. Let $\varepsilon > 0$. There exists some N such that $[-1,1]$ is covered by N disjoint open intervals (x_i, x_{i+1}) of length ε, by compactness. Let $B_i = (x_i, x_{i+1}) \times \left(-\frac{1}{2N}, \frac{1}{2N}\right)$ be corresponding open boxes. Its clear then that $L_x \subset \bigcup_{i=1}^N B_i$, and moreover $\mathrm{vol}(B_i) = \frac{\varepsilon}{N}$ and thus $\sum_{i=1}^N \mathrm{vol}(B_i) = \varepsilon$. This implies L_x has content zero. It follows identically that L_y has content zero, which completes the proof. ■

Question 4.4: *Find the absolute minimum and maximum values of $f(x_1, \ldots, x_n) = x_1^3 + \cdots + x_n^3$ in the closed unit ball $\overline{B_1(0)} \subset \mathbb{R}^n$.*

Solution. *(First Solution)* Notice by triangle inequality $|f(x)| \leq \sum_{i=1}^n x_i^2 |x_i| \leq 1$ over $\overline{B_1(0)}$ so $-1 \leq f(x) \leq 1$, with $f(-1,0,\ldots,0) = -1$ and $f(1,0,\ldots,0) = 1$ which means the absolute min, max of f over $\overline{B_1(0)}$ are $-1, 1$ respectively. ■

Solution. *(Second Solution)* We use Lagrange multipliers. Let $g(x) = x_1^2 + \cdots + x_n^2 - 1$. We compute $\nabla g(x) = 2x$ and $\nabla f = 3\left(x_1^2, \ldots, x_n^2\right)$. Setting $\nabla f(x) = \lambda \nabla g(x)$, we see that $2x_i = 3\lambda x_i^2$ for all i. Let $M = \{1 \leq i \leq n : x_i \neq 0\}$. Note that not all x_i's can be zero so $\#M \geq 1$, which means, for all $x_i \neq 0$, we find $x_i = \frac{2}{3\lambda}$. This implies

$$1 = \sum_{i=1}^{n} x_i^2 = \frac{4}{9\lambda^2} \cdot \#M \Rightarrow \lambda = \pm \frac{1}{\sqrt{\#M}} \frac{3}{2}$$

which implies

$$f(x) = \sum_{i=1}^{n} x_i^3 = \sum_{i \in M} \frac{8}{27\lambda^3} = \pm \frac{\#M}{(\#M)^{3/2}} \frac{8}{27} \frac{27}{8} = \pm \frac{1}{\sqrt{\#M}}.$$

It's clear that this is maximized by minimizing the size of M (which must have size at least 1) and taking the positive root (choosing λ positive), and minimized by minimizing the size of M as well, but taking the negative root. Thus if f maximized, minimized on $\partial B_1(0)$, it follows that $f(x) = 1, -1$ respectively. On the other hand, if $x \in B_1(0)$, $0 = \nabla f(x) = 3\left(x_1^2, \ldots, x_n^2\right)$ implies $x_i = 0$ for all i and thus $f(x) = 0$. Thus the min, max of f over $\overline{B_1(0)}$ are achieved on the boundary, i.e. $\min f = -1, \max f = 1$. ■

Question 4.5: *Suppose $f \in C^2(\mathbb{R}^n)$ satisfies*

$$f(x) = 2 \; \forall \; \|x\| = 1, \quad \|\nabla f(x)\| = 1, \frac{\partial^2 f}{\partial x_1^2}(x) + \cdots + \frac{\partial^2 f}{\partial x_n^2}(x) > 0 \; \forall \; x \in B_1(0).$$

Prove that $1 \leq f(x) \leq 2$ for all $x \in \overline{B_1(0)}$.

Solution. Let $x \in \overline{B_1(0)}$ and $y \in \partial B_1(0)$ the closest point (if multiple exist, choose any) to x on the boundary; in particular $\|x - y\| \leq 1$. By mean-value theorem, there exists some $z \in B_1(0)$ such that

$$f(x) - 2 = f(x) - f(y) = \nabla f(z) \cdot (x - y) \geq -\|\nabla f(z)\|\|x - y\| \geq -1 \Rightarrow f(x) \geq 1.$$

This proves the first inequality. For the second, we know the inequality is satisfied on the boundary. Suppose it fails inside $B_1(0)$. This implies, by compactness and continuity of f, that $\max_{\overline{B_1(0)}} f(x) > 2$ and moreover this max is achieved at some $x \in B_1(0)$. Being a maximum, this implies $\nabla^2 f(x)$ negative semidefinite, so in particular its eigenvalues are negative hence

$$0 \geq \sum \text{eigenvalues of } \nabla^2 f(x) = \operatorname{tr}\left(\nabla^2 f(x)\right) = \frac{\partial^2 f}{\partial x_1^2}(x) + \cdots + \frac{\partial^2 f}{\partial x_n^2}(x).$$

But by assumption, the quantity on the right is strictly positive, which is a contradiction. Thus f cannot attain its maximum inside $B_1(0)$ and thus $f(x) \leq 2$ everywhere. This completes the proof. ■